DO STRUCTURAL BOUNDARY TOKENS DELAY THE ACCURACY CLIFF? SCAFFOLDING, DEPENDENCY DISRUPTION, AND THE LIMITS OF A SINGLE-PASS INTERVENTION

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ABSTRACT

Large language models exhibit a sharp accuracy cliff on deterministic, multi-step sequence tasks: performance holds near ceiling over short inputs and then collapses well before any obvious capacity limit. Prior work models this collapse with a Sherrington–Kirkpatrick (SK) error-accumulation framework in which an accumulation factor $\alpha > 1$ produces runaway compounding of mistakes. We study a single-pass mitigation: inserting periodic structural boundary tokens (|) into the input, with instructions to copy them through unchanged. Across nine synthetic tasks organized into a dependency taxonomy (position-local; one-step dependency; full-prefix dependency), six models spanning three scales of one family (Qwen2.5-7B/14B/32B), two additional open architectures (Mistral-7B-v0.3, Phi-3.5-mini), and one closed frontier model (Gemini), we find that the intervention's effect is governed by two separable conditions. First, boundaries scaffold competence the model already has: four of five position-local tasks improve substantially and significantly, with the benefit growing with scale within the Qwen family, while the one position-local task on which every model sits at chance — a two-hop composed substitution that behaves as a scale-invariant capability wall from 3.8B to 32B — is made significantly *worse* by boundaries. Second, boundaries actively disrupt cross-position dependencies they land on: two purpose-built anchor tasks with trivial arithmetic but a genuine dependency (`copy_previous`, `running_max`) show significant harm that scales with boundary density and, for the full-prefix task, with sequence length. A causal ablation confirms the mechanism: rebuilding each dependency task so the recurrence genuinely resets at every boundary makes the harm shrink, vanish, or reverse into a significant benefit for every one-step task, while the full-prefix task — whose dependency can only be subdivided, never severed — is hurt substantially more, a magnitude we show is entangled with the model's own weak compliance with the reset instruction on that specific task (Section 7). Dose–response is quadratic, not monotonic: in all 43 task×model aggregate comparisons a parabola beats both a line and a saturating exponential, with an interior sweet-spot density for long sequences (per-curve robustness is scoped explicitly in the text). Beneath the accuracy results, boundaries impose a universal well-formedness tax (format-validity drops under boundaries in 9 of 9 tasks, even where accuracy rises), and the full pattern replicates on an external benchmark (CLRS-Text) and on a closed-source model. Mechanistically, boundary tokens act as measurable attention sinks ($R^2 \geq 0.97$ across models); attention entropy rises in 360 of 360 sampled trials; and zeroing boundary attention for the 10 most-boundary-attending heads erases up to 62.6% of the benefit, with head-, position-, and dose-specificity established by matched controls. The honest claim is therefore not "boundaries delay the accuracy cliff" but the more constrained: boundaries are scaffolding for a computation the model can already partly perform, they carry a real interior-optimum dose–response, and they backfire in proportion to how often they intersect a dependency the task still needs.

1 INTRODUCTION

Ask a language model to perform a simple, fully deterministic transformation — shift every character forward in a fixed alphabet, apply a substitution cipher, carry a running maximum — and accuracy does not fade smoothly as the input grows: it holds near ceiling, then falls off a cliff. Recent work (Hou et al., 2025) models this with a Sherrington–Kirkpatrick spin-glass framework in which the whole-sequence success rate follows

$$\mathrm{SAR}(N) \;=\; \exp\!\left(-\beta_0 \cdot N \cdot \alpha^{\,N-1}\right), \tag{1}$$

where β_0 is an intrinsic per-token error rate and α an error-accumulation factor: when $\alpha > 1$, each error raises the next error's probability, producing a cliff at sequence length N^*.

We test a single-pass mitigation: periodically inserting a structural marker token (|) with instructions to copy it through unchanged, interrupting attention-driven error cascades without the multiple model calls true divide-and-conquer requires. An earlier phase of this project framed the outcome as a binary locality principle — boundaries help iff each output position is independent. The expanded study here (nine tasks, three model scales in one family, three architectures, one closed model) shows that framing was incomplete in three ways.

First, **helping is conditional on competence, not only locality**: boundaries reliably improve position-local transformations, including ones built to remove any algebraic shortcut, but significantly *hurt* the one position-local task every model from 3.8B to 32B sits at or below chance on — scaffolding requires something to scaffold. Second, **harm on dependency-bearing tasks is caused by the dependency being cut, shown causally**: two anchor tasks isolate the dependency from arithmetic difficulty, and rebuilding each so the recurrence genuinely resets at every boundary makes the harm shrink, vanish, or flip to benefit for every one-step task, and substantially worsen for the full-prefix task whose dependency can only be subdivided — a magnitude a direct compliance audit qualifies (Section 7). Third, **the dose–response is quadratic, not monotonic**: across all 43 task×model curves, a parabola outperforms both a line and a saturating exponential, with the fitted optimum for long sequences sitting strictly inside the tested range.

The mechanistic program supports this: boundary tokens become measurable attention sinks, entropy rises under boundaries with no exceptions in 360 sampled trials, and causal head-level ablation with matched controls shows a small identifiable head set is load-bearing for the benefit. We also document a previously invisible well-formedness tax and replicate the central dependency split on CLRS-Text and a closed model (Gemini), where every position-local task helps at $> 5\sigma$ and every dependency-bearing task is flat or hurt.

1.1 RELATED WORK

Attention sinks. Sink behavior was first characterized by Xiao et al. (Xiao et al., 2023), extended to delimiter tokens (Chen et al., 2024), surveyed (Gu et al., 2024; Su et al., 2026), and given a mechanistic "catch, tag, release" account (Zhang et al., 2025); tabular structuring shows an analogous non-delimiter effect (Oh et al., 2024), and SinkLoRA (Zhang, 2024) operationalizes sinks at training time — the trained counterpart of our untrained markers. **Head taxonomy.** Induction heads (Olsson et al., 2022) and causal head gating (Nam et al., 2025) motivate our head-taxonomy test (Section 4.3) and ablation methodology. **Long-horizon failure.** Lost in the middle (Liu et al., 2023a), a 300-paper survey (Anand et al., 2026), HORIZON (Wang et al., 2026), LEAD (Pushkin & Abbe, 2026), and a critic-guided study (Sharif & Rehman, 2026) motivate our decomposition comparison (Section 5). **Decomposition and state-carrying.** Least-to-most (Zhou et al., 2022), program-of-thoughts (Chen et al., 2022), scratchpads (Nye et al., 2021), Transformer-XL (Dai et al., 2019), RMT (Bulatov et al., 2022), Landmark Attention (Mohtashami & Jaggi, 2023), digit-reversal training (Duan et al., 2023), position/representation injection (Nogueira et al., 2021; Shen et al., 2023), H-Net (Hwang et al., 2025), and MEMENTO (Kontonis et al., 2026) bracket the state-carrying spectrum our content-free markers sit at the cheap end of. **Benchmarks.** Tasks are generated fresh, per BeyondBench (Srivastava et al., 2025); our external check uses CLRS-Text (Markeeva et al., 2024). (Full discussion: Appendix A.1.)

2 METHODS

2.1 TASK SUITE AND DEPENDENCY TAXONOMY

Every task maps an input string of length N to an output string of the same length under a deterministic rule stated fully in the prompt. Tasks are organized by the cross-position dependency their rule introduces (Table 1).

Task	**Rule (output at position i)**	**Dependency**
Tier 1 — position-local (window 0)		
`cyclic_shift`	next character in alphabet, wrapping	none
`character_substitution`	fixed regular lookup table	none
`random_permutation`	arbitrary derangement (verified non-rotational)	none
`conditional_shift`	shift forward if even alphabet position, else back	none
`composed_permutation`	two arbitrary lookups chained	none
Tier 2 — one-step dependency (window 1)		
`running_xor`	$x_i \oplus x_{i-1}$ (binary)	one step
`running_sum_local`	$(x_i + x_{i-1}) \bmod M$, $M \in \{2, 4, 8\}$	one step
`copy_previous`	x_{i-1} (pure copy, zero arithmetic)	one step (anchor)
Tier 3 — full-prefix dependency (window ∞)		
`running_max`	$\max(x_0, \ldots, x_i)$ (trivial arithmetic)	full prefix (anchor)
running-sum-mod-M (global)	$\left(\sum_{j \le i} x_j\right) \bmod M$	full prefix
Direction-misaligned		
integer addition	digit of $A + B$; carry propagates opposite to generation	carry chain

Table 1: Task suite. The two *anchor* tasks isolate dependency structure from arithmetic difficulty (`copy_previous`: Tier 2's dependency, zero computation; `running_max`: Tier 3's shape, easiest bookkeeping). Three locality-probe tasks and one exploratory hybrid appear where used (Section 3.6, Appendix J).

Within Tier 1 the five tasks form a difficulty ladder by construction, from a stated formula (`cyclic_shift`) through a regular lookup table (`character_substitution`), a genuinely arbitrary derangement verified non-rotational (`random_permutation`), a position-conditional rule (`conditional_shift`), to two chained arbitrary lookups (`composed_permutation`) — the hardest task by a wide margin and the study's capability-wall case. Full prompt templates: Appendix C.

2.2 THE BOUNDARY INTERVENTION

In the *Base* condition the model receives the raw string. In the *Cellular_Bounds* condition a `|` marker is inserted every k characters into both the input and the expected output; the model is instructed that the marker is structural and must be copied through unchanged. Boundary density is reported as $1/k$ (0 for Base). Ground truth is computed on the full continuous sequence first and markers are spliced in afterward, so the true dependency is never removed by the intervention itself, only interrupted (Section 3.5 exploits this).

A tokenization confound, bounded directly. Digit alphabets tokenize one-token-per-character on all five models; letter alphabets do not (Base ratios 0.33–0.58, since same-case letters merge into multi-character tokens boundaries mechanically break apart), opening a non-attention channel for the five letter-alphabet tasks. Rerunning `cyclic_shift` with digit alphabets substituted for letters still shows a benefit ($+0.061$ mean TAR gap) but $2.4\times$ weaker than letters ($+0.147$), consistent with `digit_complement`'s own gap ($+0.033$): roughly 40–60% of the letter-Tier-1 benefit is genuine scaffolding, the rest tokenization — a bound, not a flag (Appendix A.2; digit-task dependency results are unaffected).

2.3 MODELS, PROMPTING, AND METRICS

Open-weight: **Qwen2.5-7B/14B/32B-Instruct** (one family, three scales), **Mistral-7B-Instruct-v0.3** (second architecture), **Phi-3.5-mini-instruct** (3.8B, third architecture); all bf16, greedy decoding, eager attention for mechanistic analyses. Closed: **Gemini** (`gemini-3.5-flash`), full nine-task grid, 3 trials/condition. Every open model is evaluated through its documented chat template throughout, which we verified materially and non-uniformly changes results if skipped (Appendix A.2).

Accuracy is TAR (per-character) and SAR (whole-sequence exact match), boundary characters stripped before scoring. Because these answer only "was the output right," not "did the model attempt the task sanely," we decompose failures with a well-formedness suite: VVR (vocabulary validity), CR (completion rate), WFR (trial-level AND of both), NEO (normalized error onset), and TAR|WF (TAR restricted to well-formed trials). Overlong rate and boundary fidelity rate exist at pilot scale only (Section 3.8).

2.4 STATISTICAL METHODOLOGY

Every condition is run for 20 trials per batch; error bars are SEM, and significance is $\Delta/\mathrm{SEM}_{\mathrm{pooled}}$ with $1\sigma/2\sigma/3\sigma$ bands. We additionally apply Benjamini–Hochberg FDR control at $q{=}0.05$ within each logical test family (main grid, reset-variant grid, and the three pooled tables): 350 of 362 uncorrected-significant main-grid cells survive (560 tests), with the 12 losses all marginal and non-load-bearing; the other families survive in full (details and $k{=}3$-inclusion sensitivity in Appendix A.4). An independent second batch is required before an effect is labeled *verified* (matrix in Appendix I); $k{=}3$ is excluded from pooled analyses as it often coincides with a task's own period.

SK parameters (β_0, α) are fit to SAR by bounded nonlinear least squares, reported only when non-degenerate, with bound-pinned fits flagged uninformative and uncertainty from a 500-iteration bootstrap. We trust an SK fit only when SAR is non-degenerate at ≥ 4 lengths, α does not pin a bound, and the bootstrap CI excludes a bound — true only for the easiest position-local grid (2-symbol `cyclic_shift`); everywhere else TAR and direct dose–response curve comparison (Section 3.2) carry the analysis. Full SK fits, bootstrap CIs, and model-selection checks are in Appendix F.

3 BEHAVIORAL RESULTS

3.1 TIER 1: BOUNDARIES SCAFFOLD EXISTING COMPETENCE

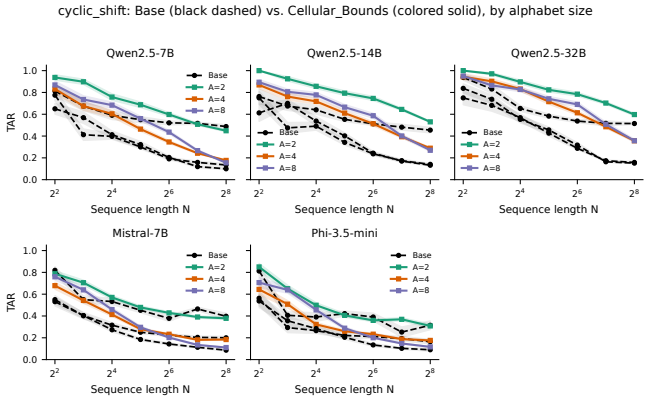


Figure 1: `cyclic_shift`, TAR vs. sequence length, all alphabet sizes, one panel per model; benefit grows with scale on Qwen, narrows on Mistral/Phi (Section 3.9).

Four of the five position-local tasks improve substantially and consistently under boundaries, across every model tested: **`cyclic_shift`** helps at every scale, growing with model size (7B→14B→32B); **`character_substitution`** helps strongly on all three Qwen sizes (+0.12 to +0.34 TAR), fading toward $N{=}64$ on Mistral/Phi (Section 3.9); **`random_permutation`**, built to remove any algebraic shortcut, helps even more cleanly despite being measurably harder (32B:

every cell positive, $+0.21$ to $+0.29$ TAR); and **`conditional_shift`** helps at 14B/32B throughout, with a single 7B/Phi ceiling-effect exception at alphabet 2. **`composed_permutation`** is the deliberate exception that proves the rule: Base sits at chance and *never improves with scale* (7B/14B/32B all 0.18–0.30; Phi-3.5-mini and Gemini bracket the same wall from below and above, Section 3.10), and boundaries make it significantly *worse* (9–10 of 10 cells negative at 7B/32B) — with no dependency anywhere in Tier 1, there is nothing for a boundary to disrupt, so **once a task exceeds what the model can do at all, there is nothing left to scaffold**. This is comprehension failure, not execution failure: well-formedness stays high (WFR 0.870 Base, 0.689 Bounds) while TAR sits at chance (Section 3.8).

The benefit is not a property of any particular marker, placement, or prompt wording: identity is fungible among out-of-vocabulary symbols, but placement *regularity* is essential (random scatter and even small jitter mostly erase the benefit) and *density*, not raw marker count, is the operative variable; the effect replicates across three independently worded prompt templates (full ablation grids in Appendix D).

(Full per-task numeric detail, scale/architecture breakdowns, and the marker-property ablation summary: Appendix A, §A.5.)

3.2 Dose–response is quadratic: an interior sweet spot

A saturating exponential fits *worse* than a flat line at the mean (R^2 negative in nearly every panel), because several long-sequence conditions peak at moderate density and decline — a monotonic family cannot represent a peak under any parameterization, a categorical failure, not a matter of degree. Fitting every boundary-density curve in the study (43 task×model combinations) three ways — line, quadratic, saturating exponential — **the quadratic wins all 43 of 43 aggregate rows** (mean R^2=0.76 vs. 0.50 linear; exponential worst everywhere, down to $R^2 = -154.7$; Figures 3–4, Appendix B).

This is an aggregate, family-level claim, not curve-by-curve robustness: fitting all 348 individual curves separately, AIC/BIC prefer the quadratic 67–70% of the time, but leave-one-density-out cross-validation prefers it over a line on only 17.5% of curves — most individual curves have too few density points (4–15) for a held-out test to distinguish a two- from three-parameter family. The correct scope is therefore that the quadratic is the right *shape family* (categorically better than exponential; better than linear in aggregate and by information criteria), while individual fitted vertices carry real per-curve uncertainty — only 2 of 43 groups retain enough CV-validated curves for a vertex-vs-N regression, so practitioner density guidance (Section 6) is deliberately qualitative. Four reasons support the quadratic beyond raw R^2, including that the fitted peak migrates interpretably with N (near-densest at short lengths, moderate density by N=256) rather than sitting fixed, which is what structure looks like. Fit quality is strongest for the two most regular lookup-table tasks and weakest for the two genuine-dependency tasks. Full per-curve robustness numbers and the 43-row table: Appendix G; extended reasoning: Appendix A, §A.6.

3.3 The anchors: dependency disruption with competence held constant

The Tier-2/3 arithmetic tasks are hard enough that parts of their Base performance sit near chance — which `composed_permutation` proved is a real confound. Two anchor tasks strip out the arithmetic entirely and keep only the dependency (Figure 5, Appendix B). **`copy_previous`** (zero computation) shows real but modest 7B competence (0.61–0.62 TAR at $N=4/8$); dense boundaries ($k \leq 3$) hurt significantly at *every* length, including where genuine skill exists to disrupt (up to -0.27 TAR, $***$), fading at sparse settings — the cleanest evidence boundaries can hurt for a reason other than "nothing to scaffold." **`running_max`** (track one value) has a high baseline throughout (0.48–1.00 TAR) and boundaries still hurt, with damage *compounding with length* (-0.075 to -0.175 by N=32–64 at 7B; -0.069 to -0.149 at 32B, whose near-ceiling Base rules out a capability artifact). Together the anchors revise the thesis: the dividing line is not difficulty but **whether the task has a cross-position dependency for a boundary to land on** — dense boundaries interrupt nearly every one-step link, sparse boundaries interrupt few, and a full-prefix dependency is crossed by every boundary, so damage compounds with length. Tier 1 never shows this because each output stands alone. (Full per-model breakdown: Appendix A, §A.7.)

3.4 HARD DEPENDENCY TASKS: TWO EFFECTS SUPERIMPOSED

`running_xor` and `running_sum_local` combine genuinely difficult arithmetic with a real one-step dependency (Figure 6, Appendix B), and decompose cleanly once a specific failure mode is measured: at 7B, roughly **38% of `running_xor` trials collapse into repeating one wrong digit** (worse under dense boundaries), a Qwen-7B-specific instability that essentially disappears at 14B (0.9%) and Mistral (0.1%). With that confound removed by scale, the picture inverts — 32B is broadly *helped* on `running_xor`, and 14B improves significantly on `running_sum_local` at $M{=}4/8$ — so on tasks combining marginal competence with a real dependency, bigger models can be net-helped by the same density that hurts a smaller model on the identical task. (Full per-modulus breakdown: Appendix A, §A.7.)

3.5 CAUSAL NECESSITY: SEVERING THE DEPENDENCY REMOVES THE HARM

Every "boundaries hurt" result above shares one construction: ground truth is computed continuously and markers spliced in after — the dependency is visually interrupted, never removed. Five *reset-variant* tasks test this: the same five tasks rebuilt (prompt *and* scoring) so the position after each boundary needs nothing from before it, with a worked example so the model genuinely attempts the reset task rather than being silently regraded.

	Original			Reset variant		
Task	Base→Bounds	Δ	Sig.	Base→Bounds	Δ	Sig.
`running_xor`	$0.411 \to 0.401$	-0.009	0.65σ	$0.448 \to 0.480$	$+0.032$	2.20σ **
`copy_previous`	$0.529 \to 0.417$	-0.112	7.38σ ***	$0.598 \to 0.546$	-0.051	3.34σ ***
`running_sum_local`	$0.309 \to 0.279$	-0.030	3.79σ ***	$0.292 \to 0.351$	$+0.059$	6.88σ ***
`reverse_xor`	$0.406 \to 0.397$	-0.009	0.67σ	$0.467 \to 0.518$	$+0.052$	3.57σ ***
`running_max`	$0.697 \to 0.658$	-0.039	2.87σ **	$0.700 \to 0.501$	-0.199	14.91σ ***

Table 2: Causal necessity ablation (Qwen2.5-7B, verified via an independent second 40-trial batch, pooled $n{=}80$; TAR pooled across lengths/densities, $k{=}3$ excluded). Severing the dependency removes or reverses the harm for every one-step task; the full-prefix task shows $\sim 5\times$ larger harm on its face, qualified by a compliance audit below.

The four one-step tasks all move the same direction: once severed, harm shrinks, vanishes, or flips to a significant benefit (`running_sum_local` fully reverses; `reverse_xor` and `running_xor` go null-to-positive; `copy_previous`'s harm halves) — direct causal support that boundaries hurt these tasks by disrupting a link the model still needs. All five variants were independently replicated in a non-overlapping second batch with no sign reversals (pooled $n{=}80$; full numbers in Appendix A, §A.8), so this result is verified, not single-batch.

`running_max` does the opposite: harm grows $\sim 5\times$ on its face. Part of this is structural — a one-step dependency can genuinely be severed (the reset position needs nothing before it), but a full-prefix dependency cannot without changing the task: **chunking does not remove the dependency, it creates more, shorter copies of it**, each still vulnerable. But this structural argument alone does not license the specific magnitude at face value, which the compliance audit below directly qualifies.

Compliance audit: does the model actually attempt the reset task? We audited this from stored generations, comparing output against both reset- and original-rule predictions at every position where they genuinely differ (48,224 positions across all five tasks; direct precedent exists — the model disobeys an unrelated echo instruction in 90–100% of trials on a different task, Appendix D). Four of five one-step tasks show real, if imperfect, compliance, favoring reset over carry-over in every case (0.385–0.475 follows-reset). **`running_max_reset` inverts this and is exactly what the $5\times$ magnitude rests on**: the model follows original carry-over behavior more than three times as often as reset semantics (0.484 vs. 0.157), *worsening with length*. This does not affect Table 2 (TAR is scored against stated ground truth regardless), but the "$\sim 5\times$" figure should be read as suggestive, not clean causal evidence: a model still substantially performing the original computation would mechanically show growing apparent harm as chunks accumulate staleness, independent of the true mechanism. The qualitative direction still holds structurally and via the audited contrast; the magnitude does not. (Full audit: Appendix A, §A.8.)

3.6 DEPENDENCY-WINDOW ALIGNMENT

Three probe tasks isolate whether boundary *placement* relative to a dependency window, not just its presence, matters. Digit complement and a Vigenère cipher both replicate the Tier-1 pattern; the Vigenère result shows alignment with a task's periodic *rule* does not matter (k=3, the key period, has no advantage over k=6). **Local block reversal** is the first task testing alignment with a genuine *dependency window* directly: edge-aligned boundaries roughly *double* accuracy, mid-window boundaries never beat Base (Table 3, Appendix A.9) — converging with the reset ablation from the placement side, and explaining why window-1 Tier-2 tasks have no alignment free lunch: every interior boundary sits on an edge regardless of placement, so for a window-1 dependency, density *is* alignment.

3.7 GLOBAL-STATE AND DIRECTION-MISALIGNED TASKS

Global running-sum-mod-M shows no boundary-adjacent recovery advantage (Figure 7, right, vs. cyclic shift's clear recovery, left) and underperforms Base at M=4/8; a *state-carrying* checkpoint (true running sum written as `[v]`) helps at the harder modulus and underperforms at the easier one — a real checkpoint helps in proportion to how much state there is to lose. The identical state-carrying design extended to `running_max` confirms this exactly (benefit rising monotonically with alphabet size, $-0.059 \rightarrow +0.108$), on a second full-prefix task built around a different operation. **Integer addition** remains the study's most uniformly reproducible negative result: Base TAR exceeds every tested boundary density at every length on all four models, since addition's right-to-left carry chain opposes generation order and every boundary splits a dependency the model is told to disregard; neither digit reversal nor an explicit scratchpad rescues it (Appendix E). (Full numbers for all of the above: Appendix A, §A.9.)

3.8 THE WELL-FORMEDNESS TAX

Scoring output validity separately from accuracy surfaces a cost accuracy alone cannot show: **WFR drops under boundaries in 9 of 9 tasks** while TAR *climbs* in all four Tier-1 helped tasks (cyclic shift: TAR 0.423 $\rightarrow$ 0.520, WFR 0.407 $\rightarrow$ 0.310) — boundaries scaffold *which* character goes where while making early stopping or invalid output somewhat more likely, a second, independent cost outweighed by the accuracy gain where TAR rises (Figure 8, Appendix B), confirmed not a 7B-only or greedy-decoding artifact by by-model and by-temperature checks. The suite also localizes specific failures invisible to TAR (a 7B-specific `copy_previous` vocabulary breakdown; a systematic wrap-around rule error in cyclic shift; the worst completion rates concentrated in the three one-step-dependency tasks) and shows the density–completion-rate correlation is negative in all 9 of 9 tasks, a real but modest secondary component of the tax. (Full by-model, by-temperature, and pilot-metric detail: Appendix A, §A.10.)

3.9 SCALE, ARCHITECTURE, AND A CAPABILITY THRESHOLD

Three regularities recur across tasks: the Tier-1 benefit *grows* with scale (7B$\rightarrow$32B) while dependency harm persists wherever headroom exists; **Mistral-7B** shows a consistent architecture signature opposite the Qwen pattern (benefit strongest short/medium, fading or reversing by long lengths, confirmed on three tasks) alongside near-zero repetition collapse; and **Phi-3.5-mini** marks a capability floor where both conditions collapse to SAR floor too fast for the decay-curve comparison to resolve. (Full breakdown: Appendix A, §A.11.)

3.10 EXTERNAL VALIDITY: CLRS-TEXT AND A CLOSED MODEL

Four external checks confirm the split is not an artifact of tasks designed for this study. **CLRS-Text** (Markeeva et al., 2024): two algorithms hard enough for 7B (Base TAR 0.16, 0.08) that neither delta is significant — the same nothing-to-scaffold pattern as `composed_permutation`. **Gemini**: the full grid on a closed model (`gemini-3.5-flash`, 3 trials/condition) replicates cleanly — position-local helps at $> 5\sigma$, dependency-bearing is hurt or flat, `composed_permutation` sits at chance too (Table 5, Appendix K; Figure 9, Appendix B). **ProcBench and string copying**: sourced from published benchmarks, ProcBench's one usable task (of 23 audited) shows a positive delta at

all 5 lengths (sign-test $p\approx0.03$); ceiling-level copying (Jelassi et al., 2024) shows a tiny null. A flip-flop Tier-2 check did not pass screening. **Vowel/consonant tagging**: a natural task reproduces both halves of the competence condition — flat at 7B (near-floor Base), helped at 32B ($+0.048$ to $+0.181$; Table 4). (Full detail: Appendix A, §A.12.)

4 Mechanism: boundary tokens as attention sinks

The mechanistic program survives the expanded behavioral picture intact: the attention-sink mechanism is *causally implicated and load-bearing for the benefit, but not sufficient for it* — firing identically whether task structure lets it translate into an accuracy gain (we stop short of claiming strict necessity, since the causal evidence is ablation on a subset of tasks/models). All mechanistic experiments use cyclic-shift with eager attention extraction; the correlational battery replicates on Qwen2.5-7B/14B/32B and Mistral-7B.

4.1 Correlational evidence

Sink scaling rises with density at every length ($R^2\geq0.98$; $R^2\geq0.97$ on replication models) and with length at fixed density (Figure 10, Appendix B). **Entropy** is higher under boundaries in **360 of 360** sampled trials, unanimous across models, including where accuracy is statistically unresolved. **Geometry**: off-diagonal mass falls, weighted distance rises slightly, and graph clustering *decreases* with density — the sink-and-entropy account survives, clustering does not (Appendix A, §A.13). Running the identical battery on `copy_previous`, where boundaries *hurt*, **the mechanism fires the same way in both directions** (same functional form, R^2=0.98–1.00; every delta metric same-signed, differing only in magnitude) — direct evidence the sink mechanism supplies the *capability* to reallocate attention while task structure decides the sign.

4.2 Causal head ablation

Zeroing boundary attention (post-softmax, renormalized) for the 10 (of 784) Qwen2.5-7B heads with highest mean boundary attention erases **2.2% / 35.2% / 33.3% / 62.6%** of Cellular's TAR benefit at N=8/16/32/64 (Figure 2, Appendix B). **Four matched controls** establish specificity: random-head, lowest-attending, and sham-position ablations produce noise-level effects; a K-sweep scales smoothly; pre-/post-softmax masking agree to 0.011 TAR; and mean-replacement (vs. zeroing), scored by log-probability, shows zero-ablation drops log-prob 3× more at N=64 — specifically *removal* of boundary attention disrupts the model. Replicates in direction on 14B/32B/Mistral-7B wherever headroom exists (Appendix A, §A.14). A parallel diagnostic on `copy_previous` finds its own top-10 heads overlap cyclic shift's persistent core at 3 of 3; ablating them *partially mitigates* the harm (0%/9.8%/9.4%/20.3% at N=8–64) — real, but a fifth of the harm versus 62.6% of the benefit under the identical procedure, evidence the harm has an additional pathway beyond attention capture alone.

4.3 What kind of heads are these?

Eight of ten boundary-attending heads rank *below* median induction-head score; nine of ten rank *above* median on positional (previous-token) attention — boundary-attending heads are positional, not content-matching, a plausible reason placement regularity matters more than marker identity (heatmaps: Figure 11, Appendix B). Rerunning identification under seven conditions shows a **persistent three-head core (layer 1/head 24, layer 3/head 24, layer 15/head 5) in every condition**, with a condition-dependent periphery (Appendix A, §A.13).

5 Cost: a cheap approximation to divide-and-conquer

True divide-and-conquer (separate model calls per 4-character chunk) wins on accuracy at every length, growing with N (TAR 0.812 → 0.871 vs. Cellular's 0.744 → 0.599 over N=8–64), but costs 3.0× Cellular (3.9× Base) in wall-clock by N=64. Boundary tokens recover a substantial fraction of that accuracy benefit for close to Base's own compute cost (throughput is density-independent;

wall-clock tracks token count, 2.6× Base at the densest setting) — a cheap single-pass approximation to a more effective multi-call strategy that, per LEAD's no-recovery bottleneck (Pushkin & Abbe, 2026), retains the shared-context recovery pathway full isolation forecloses (curves: Figure 12, Appendix B).

6 Discussion

The revised thesis. Boundaries act as scaffolding for a computation the model can already partly perform, not a source of new capability, and net effect is decided by three separable factors. (1) *Competence*: boundaries help substantially and reproducibly where genuine skill exists, increasingly with scale, and significantly *hurt* the one position-local task at chance for every model (the composed-permutation capability wall). (2) *Dependency*: with competence held constant (the anchors), boundaries hurt in proportion to how often they intersect a needed link, and the reset ablation shows causally that severing a one-step link removes or reverses the harm, while chunking a full-prefix link only subdivides it (magnitude qualified by the compliance audit, Section 7); the block-reversal alignment result makes the same point from the placement side. (3) *Dose*: response to density is quadratic with an interior optimum in all 43 aggregate comparisons — structure is good in moderation, and past the sweet spot fragmentation itself becomes the cost.

Mechanism: causally implicated, load-bearing, not sufficient. The attention-sink mechanism is real, causally load-bearing (ablation erases up to 62.6% of the benefit), and indiscriminate — firing identically whether accuracy improves, degrades, or is unresolved, and only partially mitigating harm when ablated on a harm-task (up to 20.3% vs. 62.6%), evidence disruption has a pathway beyond attention capture alone. The head-taxonomy result closes this: load-bearing heads are positional, not content-matching, precisely why periodic placement works and scattered placement fails.

Practical guidance and two hidden costs. A qualitative heuristic — densest convenient interval for $N \lesssim 32$, moderate ($k \approx 2$) for $N \gtrsim 64$ — is as far as the data supports; a quantitative vertex-vs-N formula is not (Section 3.2). The well-formedness tax and a modest compute cost are invisible to TAR alone, and nothing guarantees the accuracy gain outweighs them in stricter-format deployments. Two findings are as much about evaluation practice as about boundaries: chat-template formatting materially and non-uniformly changes results, and the dose–response analysis shows a monotonic family cannot represent a peaked phenomenon at any fit quality.

7 Limitations

- The deepest mechanistic analysis (attention-matrix figures, head taxonomy, dose–response ablation sweep) ran on Qwen2.5-7B only; the correlational core and $K=10$ ablation replicate on 14B/32B/Mistral, but the K-sweep and layer-restricted controls did not, and the battery has been extended once beyond Tier-1 (`copy_previous`), not across the full suite — the harm side remains only partially explained (Appendix A, §A.16).
- The reset-variant ablation is verified (two batches) on Qwen2.5-7B only, not other models; its compliance audit and full-prefix-magnitude caveat are reported in Section 3.5, not repeated here.
- Verification is uneven by design (Appendix I: several 14B/32B/Mistral/Phi cells single-batch, Gemini 3 trials/condition); greedy decoding throughout, though a temperature spot-check shows the effect survives sampling.
- Overlong rate, boundary fidelity, CLRS-Text, ProcBench/copying, and flip-flop are single-batch, pilot, or deferred (Section 3.10); carry-aligned addition is a confounded negative; Pauli multiplication is exploratory; CoT for `composed_permutation`, an adaptive density probe, and whether prompt engineering can recover WFR without erasing TAR gains are all future work; $k=3$ is excluded throughout.
- The tokenization confound is bounded (40–60% of letter-Tier-1 benefit is genuine scaffolding, Section 2.2; digit-task results unaffected); the quadratic dose–response is family-level, not per-curve (17.5% CV preference, Section 3.2); SK α is informative only where SAR is non-degenerate (Appendix F).

8 Conclusion

Periodic structural boundary tokens are a real, cheap, mechanistically understood single-pass intervention — but what they deliver is scaffolding, not capability. Across nine tasks, five open models spanning 3.8B–32B parameters and three architectures, and one closed frontier model, the intervention helps iff the model has genuine competence to scaffold and the task has no cross-position dependency for a boundary to land on; its benefit follows a quadratic dose–response with an interior optimum; and its harm on dependency-bearing tasks is causally attributable to the dependency being interrupted, vanishing or reversing when genuinely severed and multiplying when a full-prefix dependency can only be subdivided. Underneath sit a universal well-formedness tax, a modest compute cost, and a small positional-head-dominated attention circuit whose removal erases most of the benefit. The honest claim is therefore not "boundaries delay the accuracy cliff" but: boundaries reallocate a model's existing competence across a sequence — containing errors where positions are independent, amputating state where they are not — through a mechanism that is causally load-bearing but indifferent to whether a benefit is possible.

Reproducibility Statement

Task-generation procedures, prompt templates, and scoring conventions for all nine tasks are given in Section 2.1 and Appendix C. The statistical methodology — trial counts, the Benjamini–Hochberg FDR correction, bootstrap procedures, and the explicit criteria for when the SK fit is trustworthy — is specified in Section 2.4, with the full fitted-parameter tables in Appendix F. The causal ablation methodology, including head-selection criteria and the three matched controls, is given in Section 4.2. Model identities and versions (open-weight checkpoints and the closed-model API target) are listed in Section 2.3. Full per-condition result tables underlying every reported number are given in Appendices F–H. All code (task generation, evaluation, the eager-attention patch used for causal ablations, the well-formedness suite, and the fitting/bootstrap analysis) and the raw results underlying every reported number and figure will be made publicly available.

LLM Usage Disclosure

A large language model (Claude, Anthropic) was used substantially in the preparation of this manuscript, in ways that go beyond minor writing assistance and are disclosed here in the interest of full transparency. Specifically: (1) the manuscript's prose was extensively drafted, restructured, and iteratively revised with LLM assistance across the paper's sections, including the reorganization of an earlier internal report into the present task-taxonomy structure; (2) three figures (Figures 3, 8, and 9) were generated by LLM-authored plotting code applied to verified result data; (3) a separate LLM-based coding assistant was used to implement task generators, run experiments on GPU/SLURM infrastructure, perform statistical analyses (including the FDR correction, dose–response model selection, and the reset-instruction compliance audit reported in Section 3.5), and draft summary reports of these results, which the authors then reviewed, verified against stored data, and incorporated; (4) some follow-up experimental directions — including several of the robustness checks in Sections 3.2 and 2.2 — were suggested by the manuscript-drafting LLM in response to reviewer feedback and executed by the authors and the coding assistant. All research questions, experimental designs, data collection, and scientific conclusions were directed by the human authors, who take full responsibility for the accuracy of all claims, data, and citations in this paper; all LLM-suggested citations were independently verified against their original sources before inclusion.

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A Extended Discussion

This appendix preserves, in full, material that was condensed in the main text for space. Each subsection below corresponds to a main-text passage that now ends with a pointer here.

A.1 Extended related work

Attention sinks. Attention sinks were first characterized as a general phenomenon by Xiao et al. (Xiao et al., 2023), who found that LLMs allocate disproportionate attention to a small number of tokens regardless of semantic content, and that preserving these tokens' KV cache stabilizes streaming generation. Follow-up work extends the phenomenon to delimiter and separator tokens specifically: SepLLM (Chen et al., 2024) shows ordinary segment separators accumulate outsized attention mass and can compress a segment's information for efficient inference; a broader empirical survey (Gu et al., 2024) documents when and why sink behavior emerges over training; and a comprehensive recent survey (Su et al., 2026) organizes the literature around utilization, interpretation, and mitigation — our deliberate use of boundary tokens sits squarely in the first. A mechanistic account (Zhang et al., 2025) proposes that sink tokens catch a preceding span, tag it with a shared embedding-space direction, and release information for later retrieval — a precise vocabulary for what our boundary tokens plausibly do when they act as checkpoints rather than pure dividers. On structured non-delimiter anchors, Oh et al. (Oh et al., 2024) show tabular structuring improves factual accuracy via attention-locating effects, consistent with our finding that regular structure, not any specific symbol, carries the effect. Sink behavior has also been operationalized at training time: SinkLoRA (Zhang, 2024) fixes a small set of sink attention tokens with global attention (SF-Attn) to stabilize long-context fine-tuning — the trained counterpart of what our intervention does with untrained, prompting-time markers, and a natural point of comparison for whether learned sink placement could exceed fixed periodicity.

Mechanistic head taxonomy and head-level intervention. Olsson et al. (Olsson et al., 2022) identify induction heads implementing a pattern-completion copy operation. Our causal ablation identifies a small set of boundary-attending heads as load-bearing; Section 4.3 tests directly whether these overlap with induction heads or with purely positional (previous-token) heads. Head-level intervention as a causal tool has recently been systematized by causal head gating (Nam et al., 2025), which learns soft gates over all heads and assigns a facilitating/interfering/irrelevant taxonomy validated by ablation and mediation analysis; our diagnostic-rank-then-ablate approach is a targeted, hypothesis-driven instance of the same family, and CHG-style gating over the full head set is a natural way to extend our specificity controls beyond the single ablation mechanism used here.

Long-context and long-horizon failure. "Lost in the middle" (Liu et al., 2023a) documents position-dependent retrieval failure in long contexts, a complementary rather than competing failure mode. A 300-paper survey (Anand et al., 2026) concludes that no single reasoning paradigm dominates across settings, since outcomes hinge on how a task is posed and structured rather than on reasoning ability alone — echoing our own chat-template finding. HORIZON (Wang et al.,

2026) diagnoses long-horizon agentic breakdown at much larger scale. Most directly relevant to our decomposition comparison, LEAD (Pushkin & Abbe, 2026) identifies a no-recovery bottleneck in long-horizon execution — full isolation of steps forecloses the shared context that would let a later step correct an earlier mistake — precisely the tradeoff our boundary tokens navigate by segmenting attention without severing context. A critic-guided multi-agent study (Sharif & Rehman, 2026) similarly attributes accuracy gains to added inference-time structure rather than model size.

Decomposition and state-carrying. Least-to-most prompting (Zhou et al., 2022) and program-of-thoughts (Chen et al., 2022) address error accumulation via explicit decomposition, closer in spirit to our true divide-and-conquer baseline (Section 5) than to the single-pass intervention. Explicit scratchpads (Nye et al., 2021) make a checkpoint carry real state; Transformer-XL (Dai et al., 2019), Recurrent Memory Transformer (Bulatov et al., 2022), and Landmark Attention (Mohtashami & Jaggi, 2023) solve state-carrying architecturally; training-time digit-reversal tricks for arithmetic (Duan et al., 2023) show carry-alignment helps when trained in — our finding that the same reordering does not rescue a pretrained instruct model at prompting time (Appendix E) marks a real distinction between training-time and prompting-time fixes. A related representation-injection line makes the contrast sharper: Nogueira et al. (Nogueira et al., 2021) show that injecting explicit position-semantic tokens into the input (e.g. "3 10e1 2") lets off-the-shelf models add 60-digit numbers, and Shen et al. (Shen et al., 2023) show positional description alone reshapes arithmetic ability — content-*bearing* surface scaffolds can rescue exactly the carry-chain task our content-*free* boundaries make worse, locating the failure precisely in what the marker carries, not in adding structure per se. H-Net (Hwang et al., 2025) shows learned, data-dependent segmentation beats fixed periodic segmentation when trained jointly, and MEMENTO (Kontonis et al., 2026) trains models to manage their own context by segmenting reasoning into blocks whose state is carried forward both as compact memento text and as retained KV states — a learned, state-carrying realization of exactly the checkpoint role our state-carrying modsum redesign (Section 3.7) tests at prompting time. Our untrained periodic markers are deliberately the opposite, cheap end of that spectrum, and these learned-segmentation results jointly suggest the natural upgrade path: data-adaptive boundary placement for the Tier-1 benefit, and learned state-carrying checkpoints to mitigate the Tier-2/3 disruption we document.

Benchmarks and contamination. All tasks here are procedurally generated fresh from fixed seeds, the design rationale argued by BeyondBench (Srivastava et al., 2025) for keeping reasoning evaluation free of memorization confounds. Our external check draws two algorithms directly from CLRS-Text (Markeeva et al., 2024) using their reference generator.

A.2 Tokenization confound: full audit

In the *Base* condition the model receives the raw string. In the *Cellular_Bounds* condition a | marker is inserted every k characters into both the input and the expected output; the model is instructed that the marker is structural, must not be transformed, and must be copied through unchanged. Boundary density is reported as $1/k$ (0 for Base). Crucially, the ground truth is computed on the full continuous sequence first and markers are spliced in afterward — the true dependency is never removed by the intervention itself, only visually and tokenizationally interrupted. (Section 3.5 exploits exactly this fact.) The name "Cellular" alludes to cellular automata: periodic boundaries divide the sequence into fixed-size, locally interacting cells.

A tokenization confound specific to letter alphabets. A tokenizer audit measured in realistic prompt context (diffing tokenizations against the shared prompt prefix, avoiding isolated-character artifacts) reveals an asymmetry the study's character-level framing would otherwise hide. The digit alphabets are clean: every digit tokenizes as exactly one token on all five documented models, so the four digit-alphabet tasks (`running_max`, `running_sum_local`, `copy_previous`, `running_xor`) carry no tokenization confound. The letter alphabets are not: at Base (k=0), consecutive same-case letters routinely merge into multi-character tokens (token/character ratios of 0.33–0.58 across models; Qwen's vocabulary literally contains `ABCDEFG` as a single token), while boundary insertion mechanically breaks these merges apart — fully at k=1, partially at the moderate densities used throughout. For the five letter-alphabet tasks, the Base condition's input is therefore silently *harder to tokenize* than the boundary condition's, opening a second, non-attention channel through which denser boundaries could help: a cleaner, more atomic input representation. This

does not explain the study's central pattern — the confound-free digit tasks show the same qualitative structure, boundaries helping in some conditions and hurting in others, and the dependency-disruption results are carried by digit tasks — but it is a real partial confound for the *magnitude* of the Tier-1 letter-task benefit, disclosed here rather than left for a reader to discover (see also Limitations).

Bounding the confound directly. Rather than rest on the qualitative point that digit tasks show the same pattern, we reran the headline `cyclic_shift` task itself with digit alphabets (0–1, 0–3, 0–7) substituted for letters — identical grid, seeds, and boundary insertion, isolating the tokenization channel while holding the task fixed. Digits still benefit from boundaries ($+0.061$ mean TAR gap, positive in 15 of 21 cells), confirming the attention-scaffolding effect is real and not purely an artifact of tokenization, but the gap is roughly $2.4\times$ weaker than the letter-alphabet version of the identical task ($+0.147$) — and the size is consistent with the independently confound-free `digit_complement` task's own gap ($+0.033$; Section 3.6). Taken together, roughly 40–60% of the letter-alphabet Tier-1 benefit reported in Section 3.1 reflects genuine attention scaffolding, with the remainder attributable to the tokenization channel disclosed above — a bound on the confound's magnitude, not only a flag that it exists. This is a different, and we think cleaner, isolation than swapping to a byte-level tokenizer directly: a genuine byte-level rerun would require a model pretrained and instruction-tuned with that tokenizer, changing more than the variable under test, whereas the digit-alphabet control holds the model, task, and prompt format fixed and varies only whether the input's characters happen to merge — the confound's defining property — so we would not expect a literal byte-level tokenizer to change the estimate materially, though we have not tested one directly and flag this as the natural next step.

A.3 Chat-template formatting

Every open model is instruction-tuned and is evaluated through its documented chat template (`tokenizer.apply_chat_template(..., add_generation_prompt=True)`). We verified directly that this matters: running the full grid as raw prompt text instead materially changes results for at least one condition (the 8-symbol cyclic-shift alphabet turns from noisy and inconsistent to smooth and interpretable under the documented template), and the distortion is condition-dependent rather than uniform noise — exactly the confound that could be mistaken for an experimental effect. All results below use the chat-templated pipeline.

A.4 Statistical methodology: full detail

Every accuracy condition is run for 20 trials per batch with fresh procedurally generated sequences; error bars are standard error of the mean. Effect significance is reported as $\Delta/\mathrm{SEM}_{\mathrm{pooled}}$ with $1\sigma/2\sigma/3\sigma$ bands ($*/**/***$). To address the multiplicity these bands accumulate across the grid, we additionally apply Benjamini–Hochberg false-discovery-rate control at $q = 0.05$, corrected *within logical test families* rather than across unrelated analyses pooled together: the main nine-task grid, restricted to the five models with complete grids (Qwen2.5-7B/14B/32B, Mistral-7B, Phi-3.5-mini; every task$\times$model$\times$length$\times$axis cell, k=3 excluded as everywhere), the reset-variant grid, and the three pooled tables (causal ablation, Gemini, CLRS-Text). The results hold up well: in the main grid, 350 of the 362 cells significant at $> 2\sigma$ uncorrected survive correction (560 tests total), and the 12 that do not are all marginal edge cells (e.g. cyclic shift at N=256 on Qwen2.5-14B; `running_sum_local` at N=64 on Phi-3.5-mini) — none load-bearing for any claim in this paper. The other families survive intact: reset-variant grid 29 of 29, causal ablation 8 of 8, Gemini 6 of 6 (CLRS-Text was already null). Our verification protocol requires an independent, non-overlapping second batch of trials to reproduce an effect before it is labeled *verified*; single-batch results are labeled as such throughout, and the full verification-status matrix appears in Appendix I. Boundary density k=3 is excluded from all pooled analyses because it most often coincides with a task's alphabet or modulus period, producing degenerate boundary placements for some axis values; a full sensitivity check rerunning the FDR correction and every dose–response fit with k=3 *included* reverses no headline conclusion — every metric moves by a few percentage points without changing which side of majority/minority it sits on (largest single shift: main-grid BH survival, 350 $\rightarrow$ 330; held-out CV preference for the quadratic actually rises, 17.5% $\rightarrow$ 22.0%).

SK parameters (β_0, α) are fit to SAR by bounded nonlinear least squares ($\alpha \in [1{+}10^{-6}, 5.0]$), reported only when SAR is non-degenerate (at least two distinct values, not all zero), with fits pinned at either optimizer bound flagged as uninformative; uncertainty is quantified by a 500-iteration bootstrap over trials.

When to trust the SK fit versus TAR. Since the two frameworks answer different questions — α describes the shape of the whole-sequence collapse curve, TAR the per-position accuracy at any one length — we apply an explicit decision rule throughout. An SK fit is treated as informative only when three conditions hold: SAR is non-degenerate at four or more tested lengths, the fitted α does not pin against either optimizer bound, and the bootstrap CI excludes at least one bound. By this rule the SK framework is trustworthy for the easiest position-local conditions (the 2-symbol cyclic-shift grid, where the well-populated Cellular densities are tightly constrained — e.g. Qwen2.5-7B $k{=}1$: $\alpha = 1.037$, CI $[1.019, 1.084]$, below Base's own lower bound) and uninformative essentially everywhere else, because SAR collapses to floor before the curve has enough shape to fit. Everywhere the rule fails, TAR is the primary metric and boundary-density dose–response is characterized by direct curve comparison (linear vs. quadratic vs. saturating exponential; Section 3.2). Within the conditions that pass the rule, the SK parameters move systematically with density: fitted α falls from Base toward its lower bound as density rises across the well-constrained Cellular cells, with β_0 absorbing part of the tradeoff — i.e. where the SK lens applies at all, it tells the same story as the TAR dose–response, at the whole-sequence level. Full SK fits, bootstrap confidence intervals, model-selection (AIC) checks, and length-subset sensitivity analyses for the cyclic-shift grid are retained in Appendix F.

A.5 TIER 1: FULL PER-TASK DETAIL

Four of the five position-local tasks improve substantially and consistently under boundaries, across every model tested.

`cyclic_shift` helps at every scale, and the effect *grows* with model size across the family (7B $\rightarrow$ 14B $\rightarrow$ 32B). The only weak cells are the shortest sequences at small alphabets, where the model is already near ceiling without help.

`character_substitution` helps strongly across all three fully verified Qwen sizes, nearly every cell significant at the highest band, with independently replicated deltas of $+0.12$ to $+0.34$ TAR. Mistral and Phi-3.5-mini agree at short and medium lengths but fade — and sometimes reverse — toward $N{=}64$, an architecture signature discussed in Section 3.9.

`random_permutation` — built specifically to remove any algebraic shortcut — helps even more cleanly: TAR climbs almost monotonically with density, and this task is measurably *harder* than the two regular-rule tasks (lower Base accuracy in 60% of matched cells), so boundaries help despite the added difficulty, not because the task is easy. Qwen2.5-32B gives the tightest single result in the study: every cell positive, nearly all $***$, clustered between $+0.21$ and $+0.29$ TAR regardless of length or alphabet size.

`conditional_shift` helps at 14B and 32B across every condition (32B has no negative cells at all). At 7B one cell reverses — alphabet 2, where Base is already 0.856 — and the reversal disappears at both larger scales where the equivalent cell has headroom; Phi-3.5-mini reproduces the same alphabet-2-specific reversal. Both observations are consistent with a ceiling effect rather than a genuine exception.

`composed_permutation` is the deliberate exception that proves the rule. Base performance sits at or below random chance in nearly every condition, and — strikingly — *never improves with scale*: 7B, 14B, and 32B all sit flat at chance (0.18–0.30), with Phi-3.5-mini (3.8B) supplying a chance-level baseline from below and Gemini reproducing the wall from above (Section 3.10). This two-hop composition behaves as a structural capability wall the tested scale range does not grow out of. On this one task, boundaries make performance significantly *worse*: 9–10 of 10 cells negative at 7B and 32B (weaker at 14B, 6 of 10). With no cross-position dependency anywhere in Tier 1, there is no link for a boundary to disrupt — so this reversal isolates the first condition cleanly: **once a task exceeds what the model can do at all, there is nothing left to scaffold, and the added structure is pure overhead**.

Importantly, the model is not garbling on `composed_permutation`: its well-formedness stays high (WFR 0.870 Base, 0.689 Bounds) while TAR sits at chance — clean, well-formed, wrong guesses. The failure is comprehension, not execution (Section 3.8).

What about the boundary itself matters. The Tier-1 benefit is not a property of any particular marker or prompt wording (full ablation grids in Appendix D). Marker *identity* is fungible among symbols chosen not to occur in the task's vocabulary: `|`, `;`, `#`, and multi-character variants all clear Base with no consistent ranking among themselves, and a token-budget-matched control shows that checkpoint *frequency*, not raw token count, carries the effect. Placement *regularity* is essential, and strictly so: randomly scattered markers at the identical total count fall *below* Base at N=16 and N=32, and even *small jitter* — perturbing each periodic position by ± 1 or ± 2 characters, marker count held essentially fixed — mostly erases the benefit, with both jitter amounts falling below Base at N=64 (exact-periodic 0.615 vs. jittered 0.502/0.491 vs. Base 0.521). *Density*, not a fixed one-time count of markers, is the operative variable. Markers the model already sees constantly in ordinary text (newline, comma) are measurably weaker signals, and a genuine reserved special token is actively catastrophic, derailing generation outright. Finally, the effect is not an artifact of one instruction script: across three independently worded prompt templates (formal specification, casual instruction, terse cipher notation), the boundary condition leads in 7 of 9 template×length combinations and trails in the remaining 2 by less than 0.03, both at the shortest length only.

A.6 Dose–response: extended reasoning

An early reviewer-style question — would dropping the densest condition (k=1) and fitting a smooth saturating curve make the trend cleaner? — turns out to have an instructive answer: no. A saturating exponential $\mathrm{TAR} = \mathrm{TAR}_{\max}(1 - e^{-bx})$ fits *worse* than a flat line at the mean (R^2 negative in nearly every panel, as low as -32.7 on the restricted grid), because several long-sequence conditions do not rise monotonically at all: they peak at moderate density and decline from there. A monotonic family cannot represent a peak under any parameterization — a categorical failure, not a matter of degree. And k=1 is not an outlier breaking a clean trend; for many conditions it is one of the best-performing points.

We therefore fit every boundary-density curve in the study — 43 task×model combinations across all nine tasks and every model with enough densities to fit — three ways: straight line, quadratic, saturating exponential (aggregate fit comparison and representative raw curves in Appendix B, Figures 3–4). **The quadratic wins all 43 of 43 aggregate rows** (mean $R^2 = 0.76$ vs. 0.50 linear; the exponential is worst in every row, down to $R^2 = -154.7$).

Aggregate versus per-curve robustness. The 43-row comparison is an aggregate claim — each row averages R^2 across every alphabet/modulus×length curve for one task×model — and it should not be read as curve-by-curve robustness. Fitting all 348 individual (task, model, length, alphabet) curves separately: AIC prefers the quadratic on 234 of 348 (67%), BIC — with its stricter complexity penalty — on 245 of 348 (70%), but genuine leave-one-density-out cross-validation (refit excluding each density point, predict it held out) prefers the quadratic over the line on only 61 of 348 curves (17.5%). These results are consistent rather than contradictory: most individual curves contain only 4–15 density points, too few for a held-out test to distinguish a two- from a three-parameter family, so the quadratic's extra parameter frequently does not earn its keep out-of-sample at the single-curve level even where in-sample fit and information criteria favor it. The correctly scoped claim is therefore *family-level*: the quadratic is the right shape family for the dose–response — categorically better than the exponential everywhere, and better than the line in aggregate and by information criteria — while any individual curve's fitted parameters, including its vertex, carry per-curve uncertainty the aggregate comparison hides. Vertex-location claims below should be read against this caveat — indeed, gating vertex estimates on the CV-favors-quadratic subset leaves too little validated per-curve signal to support a general tuning formula (only 2 of 43 task×model groups retain enough CV-validated curves to fit a vertex-vs-N regression at all, both on cyclic shift), so the practitioner guidance in Section 6 is deliberately qualitative.

Four independent reasons support the quadratic beyond the R^2 table. (1) The shape matches a real mechanism: scaffolding helps at first, but past some density the fragmentation itself becomes a cost — two opposing effects produce exactly one interior peak, the same shape as any good-

in-moderation phenomenon. (2) The exponential is structurally the wrong family, as above. (3) A quadratic is the *minimum* complexity that can express one turning point; a cubic would need a mechanistic story for a second reversal, which nothing here motivates. (4) The fitted peak is a real interior feature that moves interpretably: for cyclic shift (alphabet 2, Qwen-7B) the vertex sits near the densest setting ($1/k \approx 0.87$–0.93) at N=8–32 and shifts to a moderate density ($1/k \approx 0.51$–0.64, i.e. around k=2) by N=64–256, confirmed by checking the fitted vertex directly in 8 of 9 long-sequence combinations. A peak that migrates consistently as a parameter changes is what structure, not noise, looks like.

Fit quality itself varies informatively by task: it is strongest ($R^2 > 0.9$ for every model except Phi) for the two most regular lookup-table tasks (`character_substitution`, `random_permutation`) and weakest for `running_xor` (mean 0.48) and `running_sum_local` (0.56) — precisely the tasks with genuine cross-position dependencies plus, at 7B, repetition-collapse noise (Section 3.4). A parabola cannot fix noisy data; it can only describe a real peak once one exists. The full 45-row comparison appears in Appendix G.

A.7 Anchors and hard-dependency tasks: full detail

The Tier-2/3 arithmetic tasks are hard enough that parts of their Base performance sit near chance — which `composed_permutation` proved is a real confound: a weak boundary effect might mean "task too hard," not "dependency matters." Two anchor tasks strip out the arithmetic entirely and keep only the dependency (per-model curves in Appendix B, Figure 5).

`copy_previous` ($\text{out}_i = x_{i-1}$; zero computation) shows Base competence that is real but modest at 7B (0.61–0.62 TAR at N=4/8, decaying to chance by N=16). The boundary effect is unambiguous: dense boundaries ($k \in \{1, 2, 3\}$) hurt significantly at *every* length tested — including N=4/8 where there is genuine above-chance skill to disrupt — by up to -0.27 TAR ($***$), fading to flat or slightly positive at the sparsest settings (k=8, 16). This is the cleanest single piece of evidence that boundaries can hurt for a reason other than "nothing to scaffold": real skill exists, and dense boundaries destroy it anyway. At 32B, Base is much higher (0.70–0.94) and the same moderate-density hurt persists (-0.03 to -0.04, $*$) with a U-shaped recovery at the single densest setting; 14B has the highest Base of any model (0.75–0.94) and hurts significantly at N=8–64 (up to -0.157, $***$) while showing the study's one helped cell at N=4 ($+0.062$, $***$); Mistral has the lowest, flattest Base and correspondingly the weakest, most mixed effect.

`running_max` ($\text{out}_i = \max(x_0..x_i)$; track one value) delivers a genuinely high baseline throughout (0.48–0.92 TAR at 7B, 0.96–1.00 at 32B) — and boundaries still hurt, with the damage *compounding with length*: flat-to-mixed at N=4/8/16, then strongly negative at N=32/64 across all alphabet sizes (-0.075 to -0.175, mostly $***$ at 7B; -0.069 to -0.149, all $***$, at 32B by N=64; -0.306, $***$, for Mistral at N=32, alphabet 4). More boundaries to cross means more damage to a value the model must carry the entire way, and 32B's near-ceiling Base rules out a capability artifact.

Together the anchors revise the thesis. The dividing line is not tier depth per se, and not difficulty: it is **whether the task has a cross-position dependency for a boundary to land on**. Dense boundaries interrupt nearly every one-step link (maximal damage); sparse boundaries interrupt few (damage fades toward zero); a full-prefix dependency is crossed by every boundary in the sequence, so longer sequences mean compounding damage. Tier 1 never shows any of this because each output stands alone — there, boundaries can only help.

`running_xor` and `running_sum_local` sit on top of *both* effects at once: genuinely difficult arithmetic (parts of their baselines near chance, especially at 7B) plus a real one-step dependency (per-model curves for both tasks in Figure 6, Appendix B). Their results are accordingly the messiest in the study, and decompose cleanly once a specific failure mode is measured. At 7B, roughly **38% of `running_xor` trials collapse into repeating one wrong digit**, worse under dense boundaries; the collapse essentially disappears at 14B (0.9%) and is absent for Mistral (0.1%) despite comparable parameter count — a Qwen-7B-specific instability, not a general small-model property. With that confound removed by scale, the picture inverts: at 32B, boundaries broadly *help* `running_xor` across almost every density, and at 14B `running_sum_local` improves significantly at almost every density from N=8 onward, especially at M=4 and M=8. At 7B, boundaries hurt `running_sum_local` at short lengths and turn helpful around N=32; M=2 (mathematically identical to XOR) inherits the repetition collapse, and across moduli the M=4/M=8 bound-

ary curves track each other closely while $M{=}2$ is the outlier — consistent with the collapse, not the modulus, driving its behavior. The upshot: on tasks combining marginal competence with a real dependency, the boundary effect is governed by whichever condition binds — bigger models, being both more accurate unaided and immune to the collapse, can be net-helped by the same density that hurts a smaller model on the identical task.

A.8 Reset ablation: replication and compliance audit, full detail

Every "boundaries hurt" result above shares one construction: the ground truth is computed on the continuous sequence and markers are spliced in afterward — the dependency is visually interrupted but never actually removed. That leaves a question the accuracy grids cannot answer alone: does dense-boundary harm come from breaking a link the model still needs, or is inserting | characters disruptive per se?

Five *reset-variant* tasks answer this directly: the same five dependency tasks (`running_xor`, `copy_previous`, `running_sum_local`, `reverse_xor`, `running_max`), rebuilt so the position immediately after each boundary genuinely needs nothing from before it — treated exactly like the first position of the whole sequence, with the recurrence restarting from scratch. Critically, the *prompt* is changed, not just the scoring: the model is explicitly told the rule resets at every boundary, with a worked example, so it is actually attempting the reset task rather than being silently regraded on the original one. If harm comes from breaking a needed link, severing the link should shrink, remove, or reverse it. (Numeric results: Table 2 in the main text.)

The four one-step tasks all move the same direction (Table 2): once the dependency is genuinely severed, the harm shrinks, vanishes, or flips to a significant benefit. `running_sum_local` fully reverses (hurts at 3.79σ originally, *helps* significantly reset); `reverse_xor` goes from null to significantly positive; `running_xor` moves in the same positive direction; `copy_previous`'s harm falls to under half its original size. This is direct causal support for the mechanism: boundaries hurt these tasks specifically because they disrupt a link the model still needs — remove the need, and the boundary becomes a pure helpful anchor.

Replication. All five reset variants were rerun as an independent, non-overlapping second 40-trial batch under the study's standard verification protocol. Batch 2 reproduces batch 1 in sign and magnitude for every task, with no reversals; pooled over both batches ($n{=}80$): `running_sum_local_reset` $+0.057$ (9.4σ), `reverse_xor_reset` $+0.029$ (2.8σ), `running_xor_reset` $+0.016$ (1.5σ — directionally positive but individually below the 2σ bar once pooled), `copy_previous_reset` -0.036 (3.3σ), and `running_max_reset` -0.199 (20.8σ; essentially identical across batches, -0.228 vs. -0.225) — the largest and most robust effect in the entire ablation, exactly as the subdivided-dependency account predicts. The reset-variant results are therefore verified, not single-batch.

`running_max` does the opposite — harm grows by roughly $5\times$ ($2.87\sigma \rightarrow 14.91\sigma$) on its face — and part of the reason is structural, though the paragraph immediately below qualifies the magnitude directly rather than letting it stand as clean causal evidence. A one-step dependency can genuinely be severed: the reset position needs nothing before it, full stop. A full-prefix dependency cannot be severed without changing what the task is: "reset" for `running_max` restarts the computation within each chunk, but each chunk is still its own full-prefix problem. **Chunking a full-prefix dependency does not remove the dependency; it creates more, shorter copies of it**, each still vulnerable to the same disruption — a structural argument for *why* Tier 3 should fare worse than Tier 2 under this ablation that stands on its own, independent of the specific magnitude observed. The mechanism explains Tier 2 cleanly; Tier 3 needs the different, and more qualified, story below.

Does the model actually attempt the reset task, or silently continue the original one? The causal story above assumes the reset variant is a genuinely different task the model attempts differently, not the same behavior re-scored against different ground truth — and this project has direct precedent for that assumption failing: Appendix D already shows the model disobeys an unrelated instruction (not echoing a boundary marker) in 90–100% of trials on a different task. We audited compliance directly from the stored generations: at every position immediately after a boundary where the reset-rule prediction and the original, continuous-rule prediction genuinely differ (48,224

N	Base	Aligned (k=4, window edges)	Misaligned (k=2, mid-window)
8	0.225	0.469	0.156
16	0.297	0.534	0.297
32	0.336	0.564	0.227
64	0.270	0.527	0.240

Table 3: Local block reversal: boundary placement aligned to vs. misaligned with the task's dependency window.

such positions across all five reset tasks — positions where the two predictions agree cannot distinguish compliance from coincidence), we compared the model's actual output against both.

Four of the five one-step tasks show real, if imperfect, compliance: the model follows the reset rule more often than the original carry-over rule in every case (0.385–0.475 follows-reset vs. 0.159–0.444 follows-original) — close to a coin flip for `reverse_xor_reset`/`running_xor_reset`, more clearly favored for `copy_previous_reset`/`running_sum_local_reset` — consistent with the model partially, not fully, internalizing the instruction, but favoring reset in every one-step case. **`running_max_reset` inverts this pattern, and it is exactly the result the $5\times$ magnitude above rests on**: the model follows the original carry-over behavior more than three times as often as it follows reset semantics (0.484 vs. 0.157), and compliance *degrades with length* (follows-reset falls from 0.293 at N=4 to 0.123 at N=64, while follows-original rises from 0.366 to 0.519 over the same range). This does not mean Table 2's numbers are wrong — TAR is correctly computed against the stated reset ground truth regardless of what the model actually does — but it does mean the specific "$\sim 5\times$" figure and the subdivided-dependency story built on it are harder to isolate for `running_max_reset` than for the other four tasks: a model that is substantially still performing the *original* computation would mechanically show growing apparent harm as later chunks inherit an increasingly stale running-max baseline relative to reset ground truth, independent of whether dependency-chunking is the operative mechanism. The qualitative direction — full-prefix tasks fare worse under chunking than one-step tasks do — is still supported by the structural argument above and by the contrast with the compliance-audited one-step tasks, but the specific magnitude should be read as suggestive rather than as clean causal evidence, and we report it that way rather than at face value.

A.9 Alignment and global-state tasks: full detail

Three further probe tasks (Qwen2.5-7B, 20 trials per condition) test where in a dependency a boundary lands, rather than only whether a dependency exists. **Digit complement** ($d \to 9 - d$; position-local, new domain) replicates the Tier-1 result with a density sweet spot (k=2 beats Base at every length; k=1 and k=4 trade places by length). **Vigenère cipher** (repeating 3-element key; local rule, position-dependent) is much harder overall (Base TAR 0.27 at N=8) yet every tested density beats Base at every length — and k=3, matching the key's own period, shows *no* special advantage over k=6: alignment with a task's periodic *rule* does not matter. **Local block reversal** (reverse every non-overlapping 4-character block; a genuine fixed dependency window) is the first task where boundary placement relative to a real dependency window can be tested directly, and the result is one of the cleanest in the study:

Aligned boundaries roughly double accuracy at every length; misaligned boundaries never beat Base and underperform it at three of four lengths. What matters is alignment with a genuine *dependency window*, not with periodic structure per se — converging, from the placement side, on the same conclusion the reset ablation reaches from the task side.

Why alignment has no free lunch for window-1 tasks. Local block reversal's window (4 characters) is wide enough that a periodic boundary can dodge it entirely (aligned) or land squarely inside it (misaligned) — alignment is a real choice with two outcomes. The one-step Tier-2 tasks (Section 3.3) have no such freedom: with a dependency window of exactly one position, *every* interior boundary sits directly on a dependency edge, so density alone determines how many links are cut and no placement avoids cutting some. This is exactly why the anchor tasks' density sweeps behave the way a pure exposure model predicts — harm is worst at the densest settings and fades toward the sparsest (Section 3.3) — with no separate alignment axis to tune: for a window-1 dependency, density *is* alignment, and the two collapse into the same knob.

The global cumulative tasks retained from the earlier phase complete the Tier-3 picture.

Global running-sum-mod-M ($\text{out}_i = (\sum_{j \le i} x_j) \bmod M$) shows no boundary-adjacent recovery advantage in the error-containment analysis (P(next correct | previous wrong) is identical after a boundary and elsewhere, Figure 7 in Appendix B, right), unlike cyclic shift, where post-boundary recovery is consistently higher from $N{=}8$ through $N{=}256$ (Figure 7, left), and boundaries underperform Base at every length for $M{=}4$ and $M{=}8$, with $M{=}2$ the lone exception — the same simplest-state-space pattern seen throughout. Replacing the content-free marker with a *state-carrying* checkpoint (the true running sum written as `[v]` at each boundary) helps unambiguously at the harder modulus (beats the content-free marker at every length, beats Base at 3 of 4) and underperforms at the easier one — a real checkpoint helps in proportion to how much state there is to lose track of.

State-carrying extended to the full-prefix anchor. The reset ablation (Section 3.5) showed `running_max`'s full-prefix dependency cannot be severed the way one-step dependencies can — chunking only subdivides it, which is at least part of why it showed substantially larger harm than the one-step tasks under that ablation (though see Section 3.5's compliance audit for why that specific magnitude needs care). A genuine state-carrying checkpoint is the natural alternative already validated for modsum above, so we tested the identical design on `running_max`: the true running maximum written as `[v]` at each boundary, interval 4, $N \in \{8, 16, 32, 64\}$, 20 trials per cell, across alphabet sizes 2–8. The result matches modsum's governing principle exactly: the checkpointing benefit over the content-free marker rises monotonically with alphabet size (-0.059 at size 2, $+0.064$ at size 4, $+0.108$ at size 8) — the same "helps in proportion to how much state there is to lose" pattern found for the global running sum, now confirmed on a second full-prefix task built around a different operation. Base still dominates overall, but by a much smaller margin than modsum or addition show, because `running_max`'s Base competence is already high to begin with (Section 3.3): there is less headroom for any intervention, state-carrying or not, to close a gap that starts small at low alphabet sizes and only opens up as the state to track grows.

Integer addition is retained as the designed falsification test of the causal mechanism, and it remains the most uniformly reproducible negative result in the study: at every one of 5 tested lengths on all four fully evaluated models — 20 (model, length) combinations — Base TAR exceeds *every single tested boundary density* (Qwen2.5-7B Base 0.502–0.812 vs. best-density 0.124–0.566; Qwen2.5-14B 0.688–1.000 vs. 0.141–0.875; Mistral-7B 0.462–0.975 vs. 0.115–0.672; Qwen2.5-32B 0.672–1.000 vs. 0.116–0.950). Addition's carry chain propagates right-to-left, opposite to generation order, so every boundary splits a dependency the model is told to disregard. Neither reversing digit order at prompting time nor an explicit scratchpad rescues the effect (both follow-ups on all four models; Appendix E), and a carry-aligned interleaved placement is reported as a confounded negative (the unfamiliar format prevents the model from parsing the task at all; Appendix E).

A.10 Well-formedness tax: full detail

Scoring output validity separately from accuracy surfaces a cost the accuracy numbers alone cannot show. Comparing Base to Cellular_Bounds on Qwen2.5-7B (pooled, $k{=}3$ excluded), **WFR drops under boundaries in 9 of 9 tasks** — while TAR *climbs* at the same time in all four Tier-1 helped tasks. Cyclic shift is the sharpest case: TAR rises $0.423 \rightarrow 0.520$ while WFR falls $0.407 \rightarrow 0.310$ over the identical comparison. Boundaries are doing two things at once: scaffolding which character goes where (raising TAR) while making the model somewhat more likely to stop early or emit an invalid character (lowering WFR). The headline "boundaries help Tier 1" finding is about TAR and stands; the tax is a second, independent cost sitting underneath it, outweighed by the accuracy gain exactly where TAR goes up — which is why it was invisible until VVR/CR were measured directly (Figure 8, Appendix B). The by-model breakdown confirms the tax is not a 7B quirk: WFR under boundaries sits at or below Base for most models and conditions including 14B/32B, Mistral, and Phi, with only a couple of small reversals at already-low, noisy cells.

The suite also localizes several failures invisible to TAR. `copy_previous` at 7B has a severe vocabulary breakdown (VVR 0.505 Base $\rightarrow$ 0.253 Bounds; stray letters mixed into binary output) that disappears at 14B/32B — a 7B-specific instability. Cyclic shift's low Base VVR (0.460) traces to one systematic rule error at every scale: the model ignores the wrap-around instruction and shifts the last letter past the alphabet (D$\rightarrow$E instead of D$\rightarrow$A). `running_sum_local` has the worst Base completion rate of any task (CR $= 0.403$), with `copy_previous` and `running_xor` next —

exactly the three one-step-dependency tasks, suggesting the extra bookkeeping costs completions, not just accuracy. And TAR|WF quantifies how much of a headline number is format discipline rather than comprehension: `copy_previous` Base TAR of 0.529 rises to 0.662 once malformed trials are excluded.

Does the copy-through requirement compound early termination as density rises? Breaking completion rate out by boundary interval rather than pooled Base-vs-Bounds: the per-task correlation between density and completion rate is negative — the expected denser-markers-more-early-stopping direction — in **all 9 of 9 tasks**, with modest magnitudes throughout (strongest: cyclic shift, $r = -0.41$). The compounding mechanism is universal in direction across the task suite and modest in size — a real but secondary component of the well-formedness tax.

Nor is the tax a greedy-decoding artifact: rerunning the temperature spot-check grid with the full diagnostic suite through the fixed raw-output pipeline, the negative WFR delta persists — and if anything grows slightly — under near-greedy sampling (-0.033 at T=0, -0.067 at both T=0.2 and T=0.7), while TAR's large positive delta also persists. Vocabulary validity is 1.0 at every temperature and condition, so the entire tax on this task is a completion-rate effect: the boundary condition stops or truncates slightly more often than Base, under greedy and sampled decoding alike.

Two pilot metrics motivate a rerun through the fixed pipeline (raw untruncated outputs are now always saved): on an 87-trial `running_max` verification batch, 20% of trials were *overlong*, concentrated in Base (53%) versus Bounds (11%) — consistent with boundaries constraining output length — and mean *boundary fidelity* (fraction of expected | positions reproduced exactly) was 0.75, with previously invisible failures such as a k=1 trial containing no boundary characters at all.

A.11 Scale and architecture: full detail

Three scale/architecture regularities recur across tasks. First, within the Qwen family the Tier-1 benefit *grows* with scale from 7B through 32B, while the dependency harm persists at scale wherever headroom exists (32B's near-ceiling `running_max` Base still loses -0.069 to -0.149 at N=64). Second, **Mistral-7B shows a consistent architecture signature**: in three of four verified Tier-1 tasks (cyclic shift, random permutation, conditional shift) the benefit is strongest at short and medium lengths and fades — often reversing to significant negative — by N=64–256, the opposite of the Qwen pattern; with three independent confirmations this reads as a genuine property of the architecture or training, not noise. Mistral simultaneously shows almost no repetition collapse (0.1% of `running_xor` trials vs. 38% for Qwen-7B at matched scale). Third, **Phi-3.5-mini marks a capability floor for the framework itself**: both conditions collapse to SAR floor by N=16 on the headline task, so the decay-curve comparison has too little signal to resolve — the SK machinery requires enough raw competence for both conditions' curves to be well-resolved over a non-trivial length range.

A.12 External validity: full detail

CLRS-Text. To check the findings are not an artifact of tasks built for this study, two algorithms were drawn directly from CLRS-Text (Markeeva et al., 2024) using the benchmark's own reference generator: `minimum` (running argmin over a growing prefix; Tier-3 analog) and Kadane's algorithm (best-subarray-so-far; Tier-2 analog). Because CLRS's natural alphabet is floats with multi-character output units, boundaries and scoring run at the token level here. Both tasks are genuinely hard for Qwen2.5-7B (Base TAR 0.163 and 0.076), and neither boundary delta is significant ($+0.020$, 1.06σ; -0.008, 0.72σ) — the same nothing-to-scaffold pattern `composed_permutation` establishes: boundaries can only reinforce competence that is present, and there is very little here for a 7B model to reinforce. The core distinction thus reproduces on an external benchmark's algorithms rather than only on tasks designed around it.

Gemini. The full nine-task grid was run against a closed frontier model (resolved to `gemini-3.5-flash`; 3 trials per condition, 2,040 calls, identical prompts, seeds, and scoring; per-task curves in Figure 9, Appendix B).

The pattern replicates cleanly (Table 5): every task with no cross-position dependency helps at $> 5\sigma$; `conditional_shift` and `composed_permutation` are directionally positive but non-significant — the latter because it sits at chance for Gemini too (Base 0.069), the same capability

	Qwen2.5-7B			Qwen2.5-32B		
N	Base	Best (k)	Gap	Base	Best (k)	Gap
8	0.538	0.538 (k=0)	+0.000	0.675	0.750 (k=4)	+0.075
16	0.438	0.463 (k=4)	+0.025	0.594	0.694 (k=2)	+0.100
32	0.472	0.472 (k=0)	+0.000	0.578	0.759 (k=4)	+0.181
64	0.330	0.336 (k=2)	+0.006	0.577	0.625 (k=4)	+0.048

Table 4: Vowel/consonant tagging (real position-local rule), TAR at the best boundary density per length vs. Base, 10 trials per cell. Flat at 7B (capability floor, degenerate-pattern failure mode); consistent benefit at 32B where competence exists.

wall as every open model; and every task with a real dependency is significantly hurt or flat, with `running_xor`'s null consistent with its density-dependent classification. A much stronger, closed-source model shows the same split as every open model tested — independent evidence the finding is about the mechanism, not a model family. Full per-length and per-density breakdown tables for all nine Gemini tasks, including the per-cell standard error at n=3, appear in Appendix K.

External Tier-1 tasks: ProcBench and string copying. CLRS-Text's two usable algorithms are both dependency-bearing, leaving the position-local arm of the thesis without a non-invented external check. We closed this two ways, sourcing tasks from published benchmarks rather than designing them. First, we audited all 23 task types in ProcBench (Fujisawa et al., 2024) against three criteria — string-typed, length-preserving, genuinely per-position — by inspecting real generated examples for each; exactly one survived (`task06`, "Substitute," a per-character table lookup, verified per-position with zero chaining across sampled examples), and we run it through ProcBench's own vendored reference generator at this study's lengths and seeds rather than the benchmark's fixed short-length example set. Second, plain string copying per Jelassi et al. (Jelassi et al., 2024) — the purest possible position-local task (output equals input) — at this study's alphabet sizes. Both survive the standard Base-only competence screen; results run on the full 20-trial grid. **ProcBench-Substitute**: Base TAR degrades with length as expected (0.53 at N=4 down to 0.06 at N=64), and boundaries show a positive delta at *every* tested length (5 of 5; pooled +0.044 at 1.64σ — short of 2σ on the pooled number, but five-for-five same-direction carries a sign-test $p \approx 0.03$): a second, independently sourced Tier-1 external-benchmark result in the same direction as the main study. **String copying**: Base is already at ceiling (0.97–0.98 TAR at every alphabet), and deltas are correspondingly tiny and non-significant (+0.002 to +0.007, 0.14–0.61σ) — an informative null, the nothing-to-scaffold-at-ceiling limit of the competence condition. An attempted Tier-2 external check — flip-flop language modeling (Liu et al., 2023b), reframed as per-position state transduction at two write densities — did not pass screening: the model frequently emits a single character rather than the state stream, a completion/prompt-clarity failure rather than demonstrated incapacity, so we report it as attempted-and-deferred rather than as evidence in either direction.

A real position-local task: vowel/consonant tagging. As a step beyond fully synthetic alphabets, we tested a genuinely natural, position-independent task: tag each character of a random letter string as vowel or consonant — each position's correct tag depends only on that character ($N \in \{8, 16, 32, 64\}$, $k \in \{0, 2, 4, 8\}$, 10 trials per cell, Qwen2.5-7B and 32B). The result reproduces both halves of the competence condition on a non-synthetic rule. At 7B the effect is essentially flat (best-density gaps of +0.000 to +0.025), and inspecting raw outputs shows why: Base competence is near floor (0.33–0.54 TAR), with the model frequently collapsing into a degenerate alternating `CVCVCVCV` pattern regardless of the actual input letters — the same shape of failure as `composed_permutation`'s capability wall. At 32B, where Base competence is real (0.58–0.68), boundaries help consistently and substantially at every length (+0.048 to +0.181, peaking at +0.181 at N=32). The 7B null is thus itself informative rather than a contradiction: competence-gating generalizes to a real task, not only to the synthetic ones, and the benefit appears exactly where competence does.

A.13 Mechanism: correlational, causal, and head-taxonomy detail

Sink scaling. Mean attention mass on boundary tokens — summed over all key positions landing on a | token, per query position, averaged across queries and the 20-trial batch — rises with density at every tested length with $R^2 \geq 0.98$ in every panel (Qwen2.5-7B; each panel regresses this mass

Task	Base TAR	Bounds TAR	Δ	Significance
cyclic_shift	0.659	0.852	+0.192	$5.00\sigma***$
character_substitution	0.672	0.888	+0.216	$5.29\sigma***$
random_permutation	0.606	0.846	+0.240	$5.48\sigma***$
conditional_shift	0.544	0.628	+0.084	$1.23\sigma*$
composed_permutation	0.069	0.086	+0.017	0.54σ
running_xor	0.900	0.913	+0.013	0.19σ
running_sum_local	0.936	0.803	−0.133	$5.62\sigma***$
copy_previous	0.988	0.924	−0.064	$3.37\sigma***$
running_max	1.000	0.848	−0.152	$9.47\sigma***$

Table 5: Closed-model replication (Gemini), TAR pooled across lengths/densities/alphabets, 3 trials per cell, k=3 excluded. Tasks above the rule are position-local; below, dependency-bearing.

against $1/k$ at one fixed sequence length; $R^2 \geq 0.97$ on the three replication models), and rises with length at fixed density (0.058 at N=4 to 0.177 at N=256 at the densest setting), with the growth *rate* leveling off past N=64 (full attention-mechanism panel in Figure 10, Appendix B). Chat-template special tokens act as small, flat sinks of their own (0.016–0.024 across every density and length) and do not compete with boundary tokens for a shared sink budget (density–template-mass correlation non-significant at every length).

Entropy. Attention entropy is higher under boundaries in **360 of 360** sampled trials (mean Base 1.511, Cellular 1.630), with the result replicated unanimously on all replication models — and, notably, in a harder-alphabet condition where the accuracy comparison is statistically unresolved, the mechanism still fires identically ($R^2 \geq 0.978$ sink scaling, 360/360 entropy). Whatever determines the accuracy outcome, it is not whether the mechanism fired.

Attention geometry. Under boundaries, total off-diagonal attention mass falls (0.207 vs. 0.284) while attention-weighted average distance rises slightly (0.068 vs. 0.065, in 330 of 360 trials) — less total attention off the diagonal, but some of what remains reaches specifically for boundary tokens a moderate distance back. Letter-to-letter attention dips at exactly the periodic positions boundaries occupy. One plausible-sounding hypothesis did *not* survive contact with the data: graph-theoretic clustering of the attention graph *decreases* with boundary density at every length from N=16 on, on all three original models (R^2 = 0.66–0.86, $p < 10^{-10}$) — the sink-and-entropy account is well supported; the clustering account of *why* boundaries help is not, at least as tested.

The geometry does not predict the direction: a Tier-2 anchor under the same battery. Every attention analysis above is on cyclic shift, where boundaries help. Running the identical correlational battery — matched configuration, seeds, and metrics — on copy_previous, the Tier-2 anchor where boundaries measurably *hurt*, yields a substantive qualifying finding rather than a mere replication: **the mechanism fires the same way in both directions**. Boundary-token attention mass scales linearly with density with the same clean functional form (R^2 = 0.98–1.00 at every length, both tasks), and every pooled delta metric — entropy (+0.104 vs. cyclic shift's +0.120), diagonal attention, attention-weighted distance — moves in the same direction on both tasks. The differences are of magnitude, not sign: the boundary-attention-mass slope on copy_previous is a remarkably stable 69–74% of cyclic shift's across all six lengths, and the clustering delta is the one metric larger on the harmful task (about 7×, though both are small). The attention-geometry signature alone therefore does not predict whether boundaries help or harm — the same attention capture, in the same functional form, occurs on a task where it delivers no benefit at all. This is direct Tier-2 evidence for the scoping this paper has argued throughout: the sink mechanism supplies the *capability* to reallocate attention, and it is task structure — what the redirected attention was needed for — that determines the sign of the outcome. Whatever separates help from harm lives in how the rest of the attention pattern uses the reallocated mass relative to the task's dependencies, not in whether the reallocation happens.

A.14 CAUSAL ABLATION: FULL REPLICATION AND CONTROL DETAIL

Correlation established, we test whether the circuit is load-bearing. A 5-trial diagnostic pass with output_attentions=True identifies the 10 (of 784) Qwen2.5-7B heads with the highest mean attention mass landing on boundary-token key positions — ranked by that mean, computed over all

query positions and the five diagnostic trials at the headline configuration (alphabet 2, k=2, N=16) — and the eager forward pass is patched so those heads' attention to boundary positions is zeroed post-softmax and renormalized, every other head and position untouched. Ablating 1.3% of heads erases **2.2% / 35.2% / 33.3% / 62.6%** of Cellular's TAR benefit over Base at N=8/16/32/64: at the longest length, ten heads' access to boundary tokens carries nearly two-thirds of the entire benefit, with the length-dependence mirroring the correlational mass result. (Sensitivity to *how many* heads are ablated is characterized directly below by a K-sweep; sensitivity to *which layers* the ablated heads are drawn from was not separately tested via a layer-restricted control and is noted in Limitations.)

Three matched controls establish specificity from every angle. Ablating 10 *random* heads, or the 10 *lowest*-boundary-attending heads from the same ranking, produces small, sign-inconsistent effects indistinguishable from noise; and a sham ablation of the *real* 10 heads at an equal number of random non-boundary positions likewise produces no drag. It is not the heads alone, not the positions alone, but these heads' attention to these positions that is load-bearing. A dose–response sweep over $K \in \{1, 2, 5, 10, 20, 40, 80\}$ ablated heads shows the drag scaling smoothly (15.0% of benefit erased by the single top head at N=64; 90.4% by K=20; over 100% by K=40, where the intervention stops being a targeted perturbation), concentrating the effect in well under 3% of all heads. Pre-softmax masking and post-softmax renormalization agree to within 0.011 TAR at every length, as the algebra predicts, ruling out the implementation as a confound.

A fourth control: mean-replacement, measured in log-probability. Zero-and-renormalize is a strong intervention, and a fair worry is that *any* deviation from the model's normal attention pattern — not specifically the loss of boundary attention — might produce the drag (an out-of-distribution effect). We therefore ran a gentler intervention on the identical target heads, seeds, and condition: replace each target head's attention to boundary positions with that row's own mean attention weight rather than zero, and score both interventions with a more sensitive, continuous metric — the teacher-forced log-probability of the true output — rather than binary correctness alone. The two interventions separate cleanly, and the gap grows with length: zero-ablation drops mean log-probability by $-0.118/-0.200/-0.540/-0.593$ at N=8/16/32/64, while mean-replacement — still an abnormal attention pattern, just not an absent one — moves it by only $-0.037/+0.007/-0.041/-0.186$, roughly a third of the zero-ablation drop at the longest length. It is specifically the *removal* of boundary attention that disrupts the model, not perturbation of its attention pattern per se, addressing the distribution-shift concern with a continuous measure rather than TAR alone.

The causal finding replicates in direction across models wherever headroom exists: 14B (0/0/3.8/15.8% erased, near ceiling at this density), 32B (0/0/11.4/32.0%), and Mistral-7B at k=1 — the density where its own benefit is substantial — with a clean growing drag of 22.4/28.2/34.1% through N=32 before flattening where its benefit itself has narrowed. The causal effect is genuine but naturally bounded by how much accuracy is left to erase.

Does the same circuit drive the harm, or does it only fire alongside it? Section 4.1 showed the sink mechanism fires with the same functional form on `copy_previous`, a task boundaries *hurt*; that result is correlational. We tested causally whether the load-bearing circuit is the same one responsible: a fresh diagnostic pass on `copy_previous` (Qwen2.5-7B, k=2) identifies its own top-10 boundary-attending heads, which overlap with cyclic shift's persistent three-head core (layer 1/head 24, layer 3/head 24, layer 15/head 5; Section 4.3) at 3 of 3 — and all three rank among `copy_previous`'s own top-3 heads by boundary-attention mass (0.072/0.060/0.052) — the identical core circuit is not merely present but is the single strongest boundary-attending circuit on both a help-task and a harm-task. Ablating this task-specific top-10 *partially mitigates* the harm, and the mitigation grows with length exactly as the help-side erasure does: 0% / 9.8% / 9.4% / 20.3% of the harm is erased at N=8/16/32/64. The direction is itself informative — disabling the circuit measurably reduces the damage it does, ruling out a picture where the harm is entirely independent of boundary attention — but the magnitude is the sharper result: even at the longest tested length, ablating the ten most boundary-attending heads erases only about a fifth of the harm, against up to 62.6% of the *benefit* on cyclic shift under the identical procedure and heads. The sink circuit is therefore causally implicated in both directions, but a much larger share of the benefit routes through it than of the harm — consistent with the harm having an additional pathway beyond any one head's

attention, plausibly the direct severing of the dependency itself, that the attention-sink account alone does not fully capture.

A.15 Head taxonomy and stability: full detail

Testing the 10 boundary-attending heads against the standard operational induction-head criterion, eight of ten rank *below* the population median induction score (mean percentile 30.9/100), with two above it: layer 22/head 5, which is simultaneously a top-10 boundary head and the 3rd-strongest induction head in the model (97.2nd percentile), and, more weakly, layer 21/head 6 (70.5th percentile). The mirror-image test against purely *positional* (previous-token) attention shows nine of ten ranking well *above* the median (mean percentile 83.9), with the sole exception being exactly that strong induction head (layer 22/head 5). As a population, boundary-attending heads are positional, not content-matching — consistent with a boundary's role being defined by where it sits (a fixed regular offset), and providing a plausible mechanistic reason the effect depends on placement regularity rather than marker identity (Appendix D; attention heatmaps for these three heads in Figure 11, Appendix B): a positional head's fixed-offset attention is exactly the pattern periodic placement creates and scattered placement destroys.

How stable is the identified head set? The original top-10 identification was performed once, under one condition (N=16, k=2, alphabet 2, trials 0–4). Rerunning the identical identification procedure under seven conditions — two fresh seed batches, two other lengths (N=32, 64), and two other densities (k=1, 4) — and measuring overlap against the reference set shows a graded picture: fresh seeds at the reference configuration reproduce 8 and 7 of the 10 heads (Jaccard 0.67, 0.54), while changing length or density drops overlap to 4–6 of 10 (Jaccard 0.25–0.43). Across all seven conditions, 27 distinct heads appear in some top-10 list, so roughly two-thirds of any one condition's top-10 is condition-specific — but a **persistent three-head core (layer 1/head 24, layer 3/head 24, layer 15/head 5) appears in every one of the seven conditions**, regardless of seed, length, or density. The boundary-attention circuit is therefore best described as a small stable core plus a condition-dependent periphery, consistent with the dose–response ablation's finding that most of the erasable benefit is concentrated in the first few heads.

A.16 Limitations: full original detail

- The deepest mechanistic analysis (multi-trial attention-matrix figures, head taxonomy, dose–response ablation sweep) was run on Qwen2.5-7B only; the correlational core replicates on 14B/32B/Mistral, and the K=10 causal ablation on all three, but the K-sweep was not repeated across models. Head-set stability was checked across seeds, lengths, and densities on 7B (Section 4.3), revealing a three-head persistent core with condition-dependent periphery, but not across models. The K-sweep (Section 4.2) characterizes sensitivity to *how many* heads are ablated; sensitivity to *which layers* they are drawn from — e.g. whether an equal-sized set confined to one layer reproduces the drag — was not tested. The correlational battery and the causal head ablation have each now been extended once beyond Tier-1 — to `copy_previous`, a Tier-2 anchor where boundaries hurt (Sections 4.1, 4.2) — but neither has been run across the full task suite, and the single Tier-2 causal result (harm only partially, not fully, mitigated by ablation) is itself evidence that a complete mechanistic account of the harm side remains open.
- The reset-variant causal ablation (Section 3.5) is verified via two independent batches on Qwen2.5-7B but has not been replicated on other models.
- A direct compliance audit of the reset-variant instruction (Section 3.5) finds real but imperfect compliance for four of five one-step tasks — the model favors reset semantics over silently continuing the original computation in every one-step case, though far from perfectly (0.385–0.475 follows-reset) — and a genuine inversion for `running_max_reset` specifically, where the model follows the original carry-over behavior more than three times as often as it follows reset semantics (0.484 vs. 0.157), worsening with sequence length. This does not affect any reported accuracy number — TAR is always scored against the stated ground truth regardless of what the model actually does — but it does mean the reset ablation's headline full-prefix-vs-one-step magnitude comparison, and in particular the specific "$\sim 5\times$" figure for `running_max`, should be read as suggestive of the subdivided-dependency account rather than as clean causal proof of it: a model only partially attempting the reset task would show mechanically inflated

apparent harm as chunks accumulate staleness relative to reset ground truth, independent of whether the proposed mechanism is what is actually operating. Whether this reflects the reset instruction being specifically harder to internalize for a full-prefix recurrence, or a more general difficulty following mid-generation task-switching instructions, is not resolved by this audit and is worth a dedicated follow-up.

- Verification status is uneven by design and reported transparently (Appendix I): all nine tasks are verified on Qwen2.5-7B, but several 14B/32B/Mistral cells and all Phi cells are single-batch first looks, and Gemini uses 3 trials per condition (a higher-trial rerun was deliberately not pursued; the closed-model replication should be read at that power).
- Every reported number uses greedy decoding. A targeted sampling spot-check (cyclic shift, alphabet 2, $N \in \{16, 32, 64\}$, Base vs. k=2, $T \in \{0.2, 0.7\}$, top-p=0.95, 10 trials per cell) finds the boundary benefit survives at both temperatures with direction and rough magnitude unchanged (deltas of $+0.216$ to $+0.456$ across all sampled cells, vs. $+0.198$ to $+0.456$ greedy) — so the qualitative finding is not a greedy-decoding artifact — but a full characterization under sampled deployment across tasks and models remains open.
- Overlong rate and boundary fidelity rate exist only at pilot scale — production data predates the raw-output pipeline fix and cannot be recovered retroactively.
- The carry-aligned addition placement is a confounded experiment, not a clean negative: the interleaved format prevents task parsing independent of carry alignment.
- Two natural extensions are left for future work rather than attempted here: whether chain-of-thought or tool-assisted execution lets `composed_permutation` clear its capability wall (distinguishing a single-pass limitation from a fundamental one, as the addition scratchpad follow-up did for a different task), and whether a cheap adaptive probe (two or three densities, or a meta-predictor from early attention statistics) can locate the interior-optimum density without the full grid search this paper relies on.
- CLRS-Text coverage is two algorithms on one model at a single batch, and its token-level boundary scheme differs structurally from the character-level scheme used elsewhere. The additional external Tier-1 checks (ProcBench-Substitute, string copying) are single-model (Qwen2.5-7B), the flip-flop Tier-2 external check is deferred pending a prompt-clarity fix, and whether prompt engineering (stricter output schemas, self-checks) can recover the well-formedness tax without erasing TAR gains is explicitly left as future work.
- Pauli string multiplication remains exploratory and underpowered (Appendix J); boundary density k=3 is excluded from pooled analyses throughout.
- The five letter-alphabet tasks carry a tokenization confound absent from the four digit-alphabet tasks: Base inputs tokenize into merged multi-character tokens that boundary insertion mechanically breaks apart, a second channel through which boundaries could help those tasks specifically. This is now bounded rather than only flagged: a digit-alphabet replication of `cyclic_shift` isolates the channel and finds roughly 40–60% of the letter-alphabet Tier-1 benefit is genuine attention scaffolding (Section 2.2), with the confound accounting for the remainder. The dependency-disruption results throughout the paper are carried by confound-free digit tasks and are unaffected.
- The quadratic dose–response is an aggregate, family-level claim: leave-one-density-out cross-validation prefers the quadratic over a line on only 17.5% of the 348 individual curves (most have too few density points to distinguish the families out of sample; Section 3.2), so single-curve vertex estimates — including the practitioner heuristic — should not be over-read.
- The SK α framework is informative only where SAR is non-degenerate; for most conditions beyond the easiest, TAR and the dose–response fits carry the analysis (full SK treatment in Appendix F).

B Additional Figures

Figures referenced from the main text but not reproduced there for space; each is cross-referenced at the relevant point in Sections 2–5.

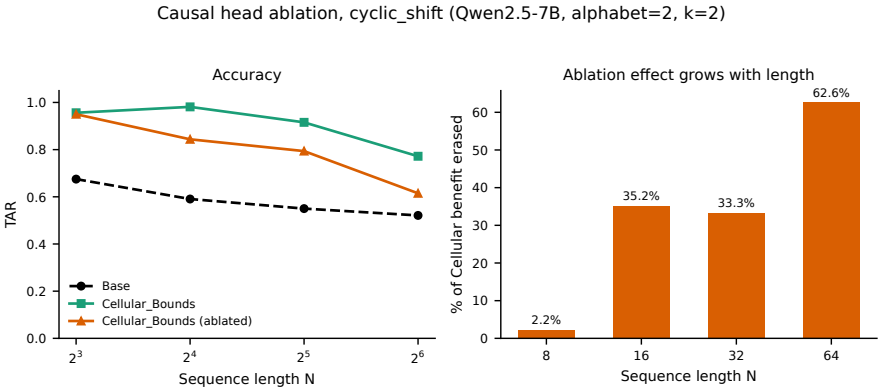


Figure 2: Causal head ablation, `cyclic_shift` (Qwen2.5-7B, alphabet 2, k=2). Left: TAR vs. sequence length for Base, normal Cellular_Bounds, and Cellular_Bounds with the top-10 boundary-attending heads ablated. Right: the percentage of Cellular's TAR benefit over Base that ablation erases, by length.

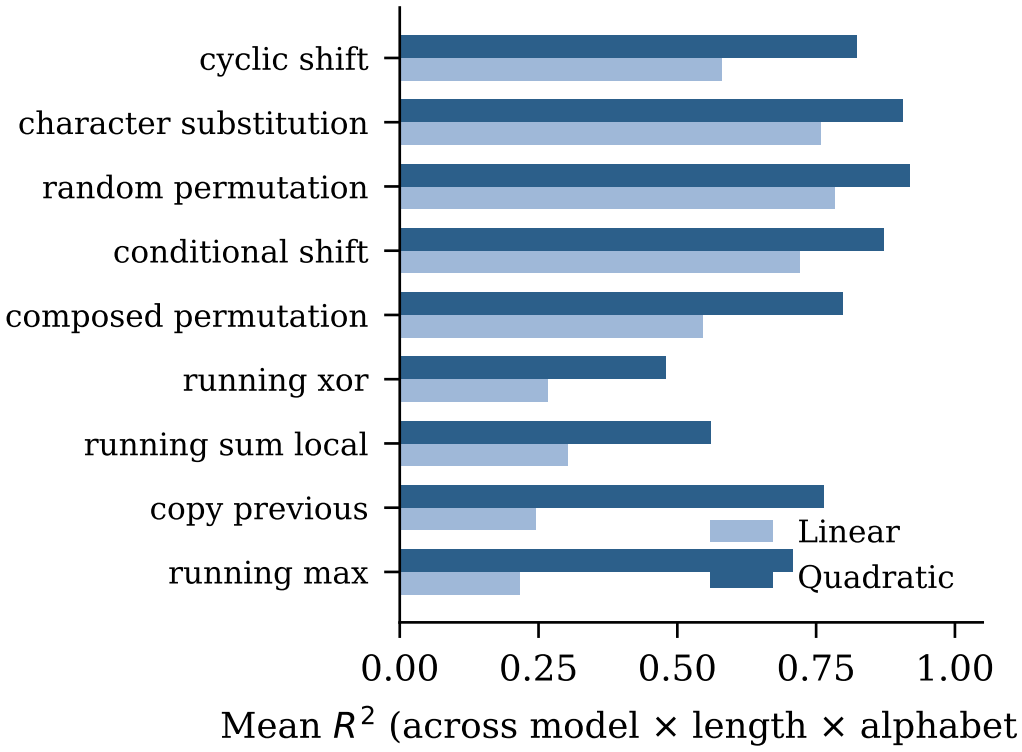


Figure 3: Mean R^2 for linear vs. quadratic fits, by task, averaged across every model×length×alphabet curve (the 43 rows of Table 9). The quadratic wins every task; the saturating exponential (omitted here for scale — see text and Appendix G) is worst everywhere. Figure 4 shows the underlying raw density-vs-TAR scatter with fitted curves and vertex markers for one representative task.

C PROMPT TEMPLATES

The headline task's template and the two paraphrase-robustness variants (Section 3.1) are reproduced below; the addition-variant prompts appear in Appendix E and the decomposition prompt

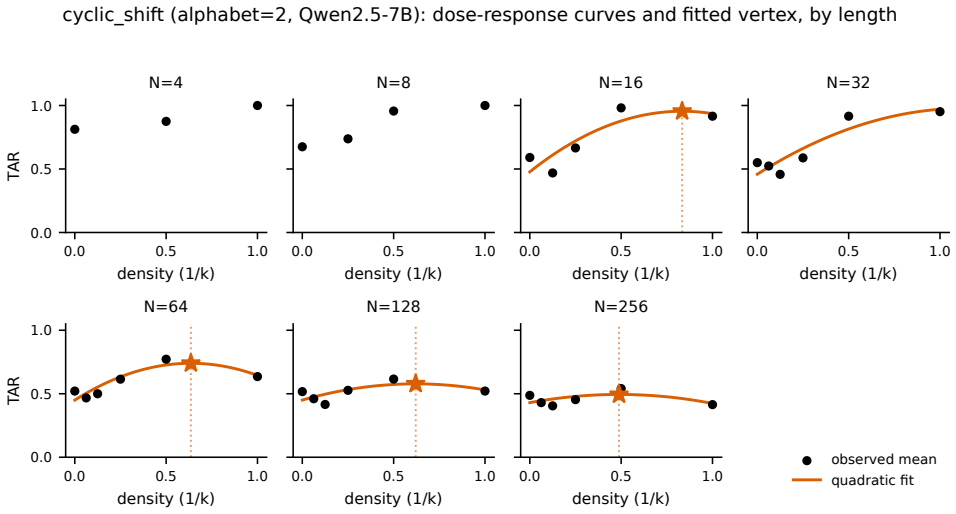


Figure 4: The raw dose–response data underlying the aggregate comparison in Figure 3: `cyclic_shift` (alphabet 2, Qwen2.5-7B), TAR vs. boundary density at each tested sequence length, with the fitted quadratic and its vertex (star) marked. No fit is attempted at $N{=}4, 8$, where TAR is too close to ceiling for curvature to be identified. The vertex sits near the densest tested setting for short-to-medium lengths and migrates toward a moderate density by $N{=}256$ — the peak-migration finding discussed in Section 3.2.

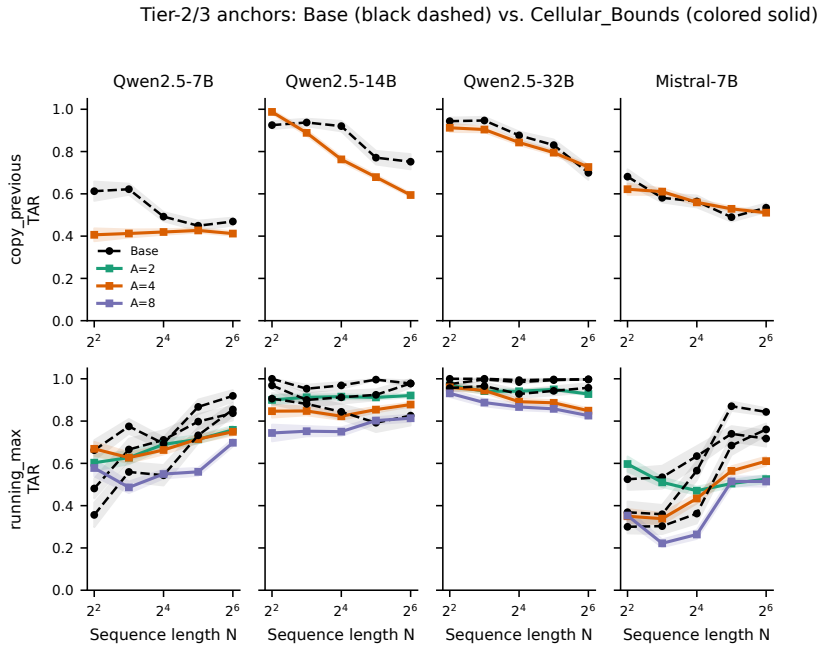


Figure 5: The two anchor tasks, TAR vs. sequence length, Base (black, dashed) vs. Cellular_Bounds (colored, one line per alphabet size), one panel per model. `copy_previous` (top row) shows dense-boundary harm even where Base competence is real (short lengths); `running_max` (bottom row) shows harm compounding with length across every model, including 32B, whose near-ceiling Base rules out a capability artifact.

in Appendix L. The eight remaining task templates and all five reset-variant original-vs-reset pairs (shown on identical inputs, equivalence verified programmatically) follow.

character_substitution (N=8, k=2, axis=4)

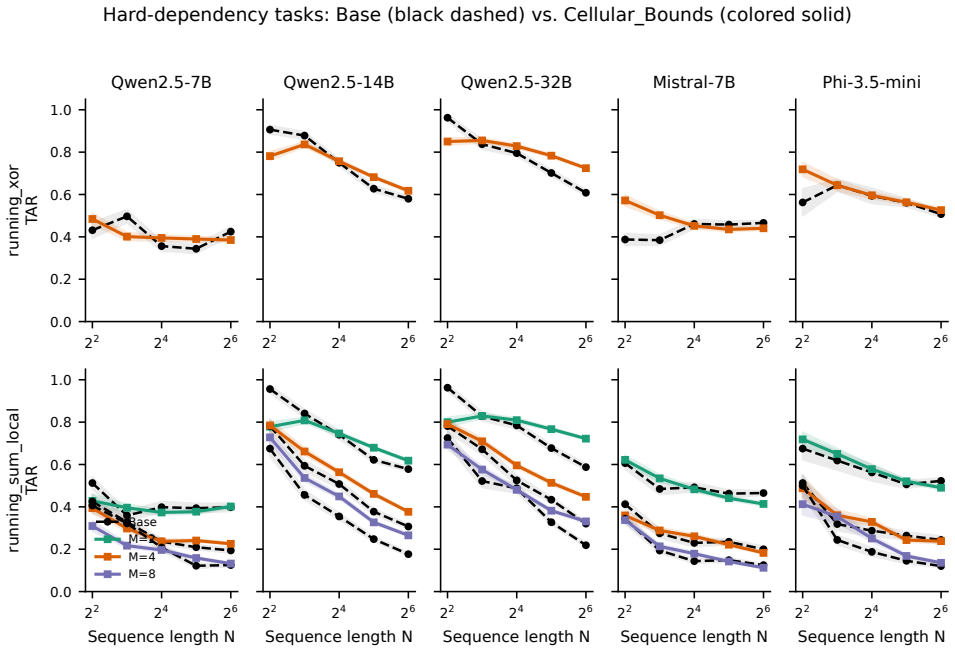


Figure 6: `running_xor` (top) and `running_sum_local` at all three moduli (bottom), TAR vs. sequence length, Base (black, dashed) vs. Cellular_Bounds (colored solid), by model. The two effects superimposed in Section 3.4 are visible directly: Qwen2.5-7B's noisy, sometimes-positive `running_xor` panel reflects the repetition-collapse confound, while 14B/32B show broadly positive boundary effects once that confound is absent.

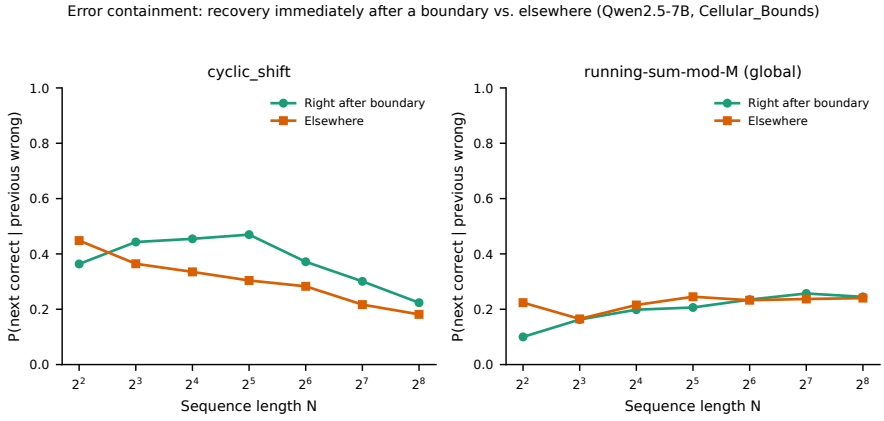


Figure 7: Error containment: P(next character correct | previous character wrong), split by whether the position immediately follows a boundary vs. elsewhere in the sequence (Qwen2.5-7B, Cellular_Bounds). `cyclic_shift` (left) shows post-boundary recovery consistently above elsewhere from $N{=}8$ through $N{=}256$ (pooled across those lengths: 0.2723 right-after-boundary vs. 0.2114 elsewhere); global running-sum-mod-M (right) shows no separation at any length — direct behavioral evidence that boundaries contain errors specifically where the task has no cross-position dependency for the error to have corrupted in the first place.

```
You are a deterministic sequence transformer.
Task: Character substitution mapping.
Alphabet: A, B, C, D
Mapping: A->C, B->D, C->A, D->B
Rule: Replace each character with its mapped character. This is a fixed
↪  one-to-one substitution cipher.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
```

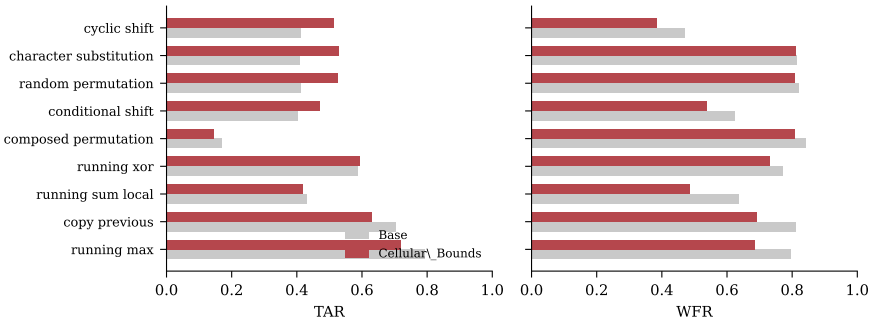


Figure 8: The well-formedness tax, pooled across all documented models per task (k=3 excluded): TAR (left) rises under Cellular_Bounds for the four helped Tier-1 tasks while WFR (right) falls for most tasks regardless of whether TAR rises or falls — the two costs move independently. Data are the 86 pooled task×model×condition rows underlying Appendix H.

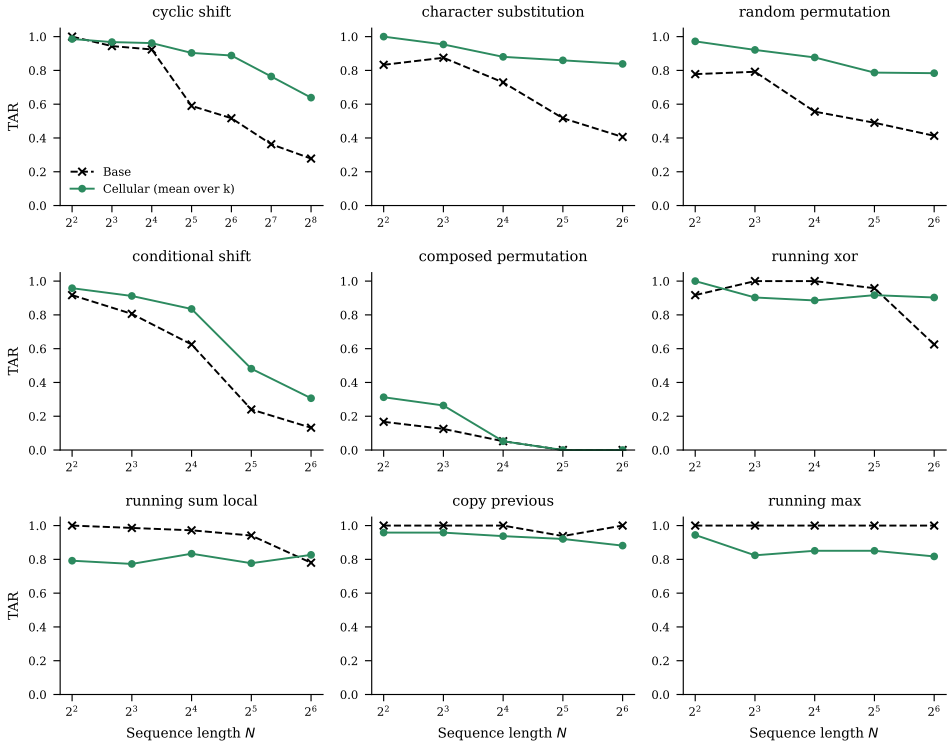


Figure 9: Gemini TAR vs. sequence length, all nine tasks, Base (black, dashed) vs. Cellular_Bounds (green, mean across tested densities). The competence/dependency split reproduces visually: Base and Cellular separate cleanly on the four helped Tier-1 tasks, track closely or invert on the dependency-bearing tasks, and both sit near floor throughout on `composed_permutation`. Data are the complete per-length, per-density cells of Appendix K (n=3 per cell).

```
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: CA|BA|CC|DA
Output:
```

random_permutation (N=8, k=2, axis=4)

```
You are a deterministic sequence transformer.
Task: Character substitution mapping.
```

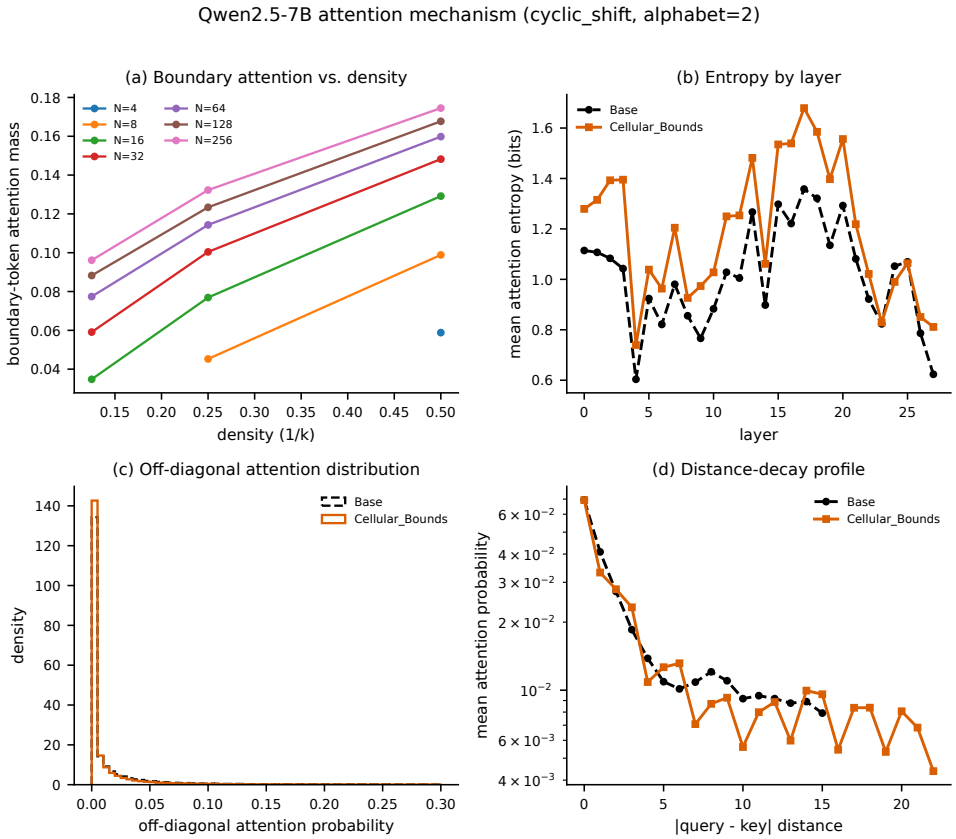


Figure 10: Qwen2.5-7B attention mechanism (`cyclic_shift`, alphabet 2): (a) boundary-token attention mass vs. density, faceted by length; (b) mean attention entropy by layer, Base vs. Cellular_Bounds; (c) off-diagonal attention distribution; (d) distance-decay profile. Panels (a)–(b) show the sink-scaling and entropy results directly; (c)–(d) show the leftward mass shift and slightly increased weighted distance discussed in the text.

```
Alphabet: A, B, C, D
Mapping: A->D, B->C, C->A, D->B
Rule: Replace each character with its mapped character. This is a fixed
↪  one-to-one substitution cipher.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: CA|BA|CC|DA
Output:
```

conditional_shift (N=8, k=2, axis=4)

```
You are a deterministic sequence transformer.
Task: Branching character shift.
Alphabet: A, B, C, D (positions numbered 0, 1, 2, ... in this order)
Rule: For each character, look at its position number in the alphabet.
- If the position is EVEN, replace the character with the NEXT character
↪  in the alphabet (e.g. A -> B).
- If the position is ODD, replace the character with the PREVIOUS
↪  character in the alphabet (e.g. B -> A).
Wrap around the alphabet if needed.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries. DO NOT shift them. Copy them exactly to the output.
```

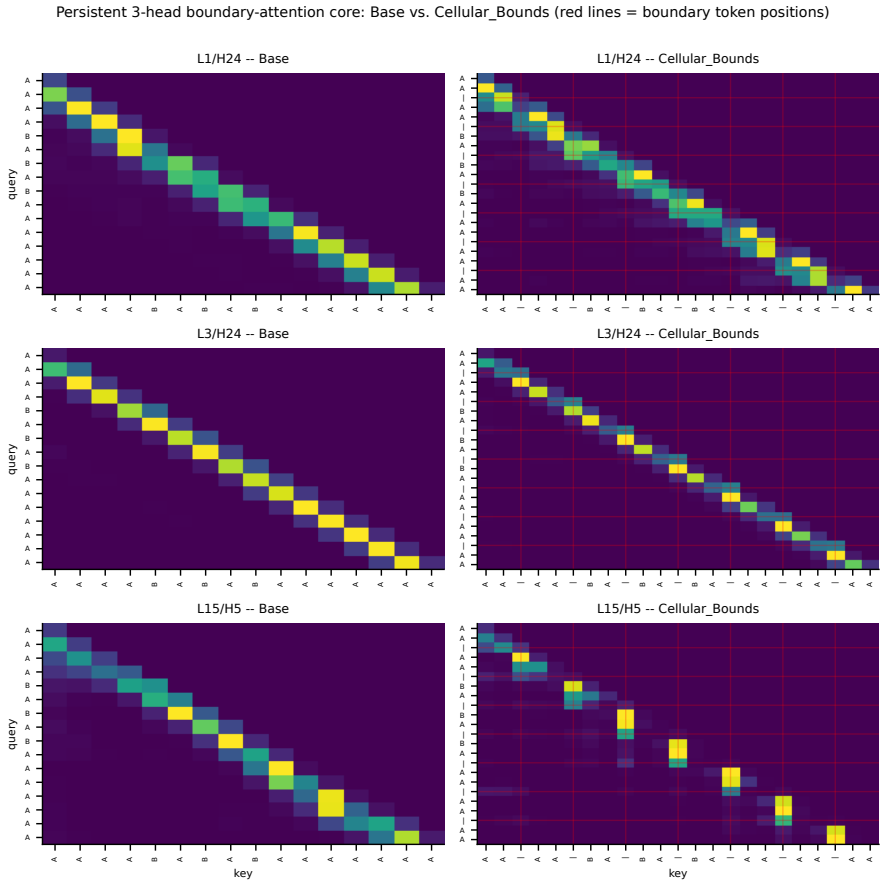


Figure 11: Attention heatmaps for the persistent three-head core (layer 1/head 24, layer 3/head 24, layer 15/head 5), Base vs. Cellular_Bounds, `cyclic_shift` (alphabet 2, N=16, k=2). All three heads show the same near-diagonal local pattern under Base; under boundaries, attention visibly pulls toward the | positions (red gridlines) exactly where the diagonal crosses them, most sharply for layer 15/head 5.

```
No spaces, no explanations.

Input: CA|BA|CC|DA
Output:
```

composed_permutation (N=8, k=2, axis=4)

```
You are a deterministic sequence transformer.
Task: Composed character substitution (apply two fixed mappings in
↪  sequence).
Alphabet: A, B, C, D
Step 1 mapping: A->B, B->A, C->D, D->C
Step 2 mapping: A->D, B->C, C->A, D->B
Rule: For each character, first apply the Step 1 mapping, then apply the
↪  Step 2 mapping to that result.
This gives the final output character. Apply this to every character
↪  independently.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: CA|BA|CC|DA
Output:
```

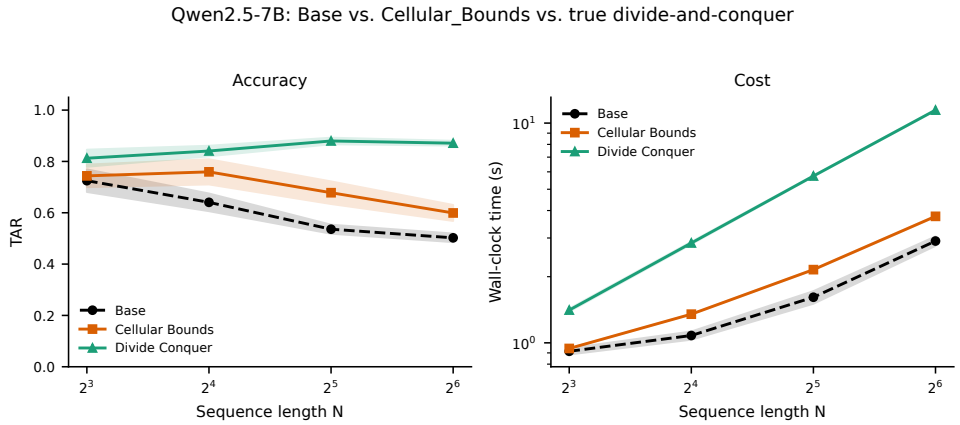


Figure 12: `cyclic_shift` (alphabet 2), Qwen2.5-7B: TAR (left) and wall-clock time per trial (right, log scale) for Base, Cellular_Bounds (single-pass boundary tokens), and true divide-and-conquer (the sequence split into separate model calls per chunk, merged after).

running_xor (N=8, k=2)

```
You are a deterministic sequence transformer.
Task: Running XOR with previous input.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the XOR of the current input digit with
↪  the previous input digit.
The first digit is copied as-is (no previous input to XOR with).
Each subsequent digit: output[i] = input[i] XOR input[i-1]
Example: Input 1011 -> Output 1110 (calculation: 1, 0XOR1=1, 1XOR0=1,
↪  1XOR1=0)
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
```

running_sum_local (N=8, k=2, axis=4)

```
You are a deterministic sequence transformer.
Task: Running sum with previous input (local window).
Alphabet: 0-3 (digits)
Modulus: 4
Rule: At each position, output the sum of the current input digit and the
↪  previous input digit, modulo 4.
The first digit is copied as-is (no previous input to add).
Each subsequent digit: output[i] = (input[i] + input[i-1]) mod 4
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|21|11|00
Output:
```

copy_previous (N=8, k=2)

```
You are a deterministic sequence transformer.
Task: Copy the previous input digit.
Alphabet: 0, 1 (binary digits)
```

```
Rule: At each position, output the PREVIOUS input digit (not the current
↪  one).
The first digit is copied as-is (no previous input to copy).
Each subsequent digit: output[i] = input[i-1]
Example: Input 1011 -> Output 1101 (calculation: 1 (copied), then 1, 0, 1
↪  -- each shifted from the position before)
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
```

running_max (N=8, k=2, axis=4)

```
You are a deterministic sequence transformer.
Task: Running maximum.
Alphabet: 0-3
Rule: At each position, output the LARGEST digit seen so far, including
↪  the current position.
Scan left to right; once a digit is output, it never decreases.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|21|11|00
Output:
```

running_xor: original vs. reset variant (N=8, k=2)

```
--- ORIGINAL ---
You are a deterministic sequence transformer.
Task: Running XOR with previous input.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the XOR of the current input digit with
↪  the previous input digit.
The first digit is copied as-is (no previous input to XOR with).
Each subsequent digit: output[i] = input[i] XOR input[i-1]
Example: Input 1011 -> Output 1110 (calculation: 1, 0XOR1=1, 1XOR0=1,
↪  1XOR1=0)
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
--- RESET ---
You are a deterministic sequence transformer.
Task: Running XOR with previous input, reset at boundaries.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the XOR of the current input digit with
↪  the previous input digit.
The first digit is copied as-is (no previous input to XOR with).
Each subsequent digit: output[i] = input[i] XOR input[i-1]
IMPORTANT: The input contains vertical bars (|), structural boundaries.
↪  Immediately after each boundary, treat that position exactly like the
↪  first digit of the whole input -- copy it as-is (nothing before a
↪  boundary counts as "previous"), then resume the XOR rule from there
↪  until the next boundary.
Example: Input 1011|0010 -> Output 1110|0011 (first block: 1 copied,
↪  0XOR1=1, 1XOR0=1, 1XOR1=0; second block restarts: 0 copied, 0XOR0=0,
↪  1XOR0=1, 0XOR1=1)
```

```
Copy the boundary characters themselves exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
```

`copy_previous`: original vs. reset variant (N=8, k=2)

```
--- ORIGINAL ---
You are a deterministic sequence transformer.
Task: Copy the previous input digit.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the PREVIOUS input digit (not the current
↪  one).
The first digit is copied as-is (no previous input to copy).
Each subsequent digit: output[i] = input[i-1]
Example: Input 1011 -> Output 1101 (calculation: 1 (copied), then 1, 0, 1
↪  -- each shifted from the position before)
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
--- RESET ---
You are a deterministic sequence transformer.
Task: Copy the previous input digit, reset at boundaries.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the PREVIOUS input digit (not the current
↪  one).
The first digit is copied as-is (no previous input to copy).
Each subsequent digit: output[i] = input[i-1]
IMPORTANT: The input contains vertical bars (|), structural boundaries.
↪  Immediately after each boundary, treat that position exactly like the
↪  first digit of the whole input -- copy IT as-is (nothing before a
↪  boundary counts as "previous"), then resume the shift-by-one rule
↪  from there until the next boundary.
Example: Input 1011|0010 -> Output 1101|0001 (first block: 1 copied, then
↪  1,0,1 shifted from the position before; second block restarts: 0
↪  copied, then 0,0,1 shifted from the position before, within that
↪  block only)
Copy the boundary characters themselves exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
```

`running_sum_local`: original vs. reset variant (N=8, k=2)

```
--- ORIGINAL ---
You are a deterministic sequence transformer.
Task: Running sum with previous input (local window).
Alphabet: 0-3 (digits)
Modulus: 4
Rule: At each position, output the sum of the current input digit and the
↪  previous input digit, modulo 4.
The first digit is copied as-is (no previous input to add).
Each subsequent digit: output[i] = (input[i] + input[i-1]) mod 4
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.
```

```
Input: 00|21|11|00
Output:
--- RESET ---
You are a deterministic sequence transformer.
Task: Running sum with previous input (local window), reset at
↪  boundaries.
Alphabet: 0-3 (digits)
Modulus: 4
Rule: At each position, output the sum of the current input digit and the
↪  previous input digit, modulo 4.
The first digit is copied as-is (no previous input to add).
Each subsequent digit: output[i] = (input[i] + input[i-1]) mod 4
IMPORTANT: The input contains vertical bars (|), structural boundaries.
↪  Immediately after each boundary, treat that position exactly like the
↪  first digit of the whole input -- copy it as-is (nothing before a
↪  boundary counts as "previous"), then resume the running-sum rule from
↪  there until the next boundary.
Copy the boundary characters themselves exactly to the output.
No spaces, no explanations.

Input: 00|21|11|00
Output:
```

`reverse_xor`: original vs. reset variant (N=8, k=2)

```
--- ORIGINAL ---
You are a deterministic sequence transformer.
Task: Running XOR with next input.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the XOR of the current input digit with
↪  the NEXT input digit.
The last digit is copied as-is (no next input to XOR with).
Each other digit: output[i] = input[i] XOR input[i+1]
Example: Input 1011 -> Output 1101 (calculation: 1XOR0=1, 0XOR1=1,
↪  1XOR1=0, 1)
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
--- RESET ---
You are a deterministic sequence transformer.
Task: Running XOR with next input, reset at boundaries.
Alphabet: 0, 1 (binary digits)
Rule: At each position, output the XOR of the current input digit with
↪  the NEXT input digit.
The last digit is copied as-is (no next input to XOR with).
Each other digit: output[i] = input[i] XOR input[i+1]
IMPORTANT: The input contains vertical bars (|), structural boundaries.
↪  Immediately BEFORE each boundary, treat that position exactly like
↪  the last digit of the whole input -- copy it as-is (nothing after a
↪  boundary counts as "next" for it), then the rule resumes normally on
↪  the other side of the boundary.
Example: Input 1011|0010 -> Output 1101|0110 (first block: 1XOR0=1,
↪  0XOR1=1, 1XOR1=0, 1 copied since it's right before the boundary;
↪  second block: 0XOR0=0, 0XOR1=1, 1XOR0=1, 0 copied since it's the last
↪  digit)
Copy the boundary characters themselves exactly to the output.
No spaces, no explanations.

Input: 00|10|00|00
Output:
```

`running_max`: original vs. reset variant (N=8, k=2)

```
--- ORIGINAL ---
You are a deterministic sequence transformer.
Task: Running maximum.
Alphabet: 0-3
Rule: At each position, output the LARGEST digit seen so far, including
↪  the current position.
Scan left to right; once a digit is output, it never decreases.
Constraint: The input contains vertical bars (|). These are structural
↪  boundaries.
DO NOT transform them. Copy them exactly to the output.
No spaces, no explanations.

Input: 00|21|11|00
Output:
--- RESET ---
You are a deterministic sequence transformer.
Task: Running maximum, reset at boundaries.
Alphabet: 0-3
Rule: At each position, output the LARGEST digit seen so far, including
↪  the current position.
Scan left to right; once a digit is output, it never decreases -- within
↪  the current block.
IMPORTANT: The input contains vertical bars (|), structural boundaries.
↪  Each boundary starts a brand new block: the running maximum RESETS to
↪  nothing at that point and starts counting fresh from the first digit
↪  of the new block, ignoring everything before the boundary.
Copy the boundary characters themselves exactly to the output.
No spaces, no explanations.

Input: 00|21|11|00
Output:
```

Original (cyclic shift).

```
You are a deterministic sequence transformer.
Task: Cyclic character shift.
Alphabet: A, B
Rule: Replace every character with the next one in the alphabet.
Mapping: B -> A, and A -> B.
Constraint: The input contains vertical bars (|). These are structural
boundaries. DO NOT shift them. Copy them exactly to the output.
No spaces, no explanations.

Input: {input}
Output:
```

Paraphrase 1.

```
You perform one job only: shifting letters forward in a fixed alphabet.
Letters in use: A, B
For every letter you see, write the next letter in this list instead
(wrapping from B back to A).
Some characters in the input are pipes (|); these are dividers, not
letters -- reproduce them unchanged in your answer.
Respond with the transformed string only. Do not add spaces or
↪  commentary.

Sequence: {input}
Result:
```

Paraphrase 2.

```
Apply a fixed substitution cipher: B -> A -> A.
In plain terms, replace each symbol with the one that follows it in A, B,
cycling back to the start after the last symbol.
Any "|" characters present must be left exactly where they are -- they
are not part of the alphabet.
Give only the resulting string, with no spaces and no explanation of
your reasoning.

>>> {input}
<<<
```

D MARKER ABLATIONS: WHAT ABOUT THE BOUNDARY MATTERS

Summary of the marker-property ablations (Qwen2.5-7B, cyclic shift): (i) *identity* is fungible among out-of-vocabulary symbols (`|`, `;`, `#`, `||`, `<|>` all clear Base with no consistent ranking); (ii) *placement regularity* is essential, and strict: randomly scattered markers at matched count fall *below* Base at $N{=}16/32$, and a jitter ablation (each periodic position perturbed by ±1 or ±2, drawn fresh per boundary per trial, cyclic shift, alphabet 2, $k{=}4$, same seeds as the other marker ablations) shows even approximate periodicity is not enough — exact-periodic beats Base at every length (e.g. 0.738 vs. 0.675 at $N{=}8$; 0.615 vs. 0.521 at $N{=}64$) while jittered placement mostly does not, with both jitter amounts *below* Base at $N{=}64$ (0.502 and 0.491); (iii) *density*, not one-time count, is the operative variable; (iv) naturally occurring tokens (newline, comma) are measurably weaker signals and can hurt at short lengths; (v) a genuine reserved special token (`<|object_ref_start|>`) is catastrophic, underperforming Base from $N{=}16$ on with generation derailed almost immediately; (vi) tokenization is controlled — all headline markers are token-count-consistent across all model tokenizers, and a token-budget-matched control shows checkpoint *frequency*, not raw token count, does the work (`<|>` at interval 12 loses to `|` at interval 4 on an equalized budget at every length); (vii) an attempted echo-vs-segmentation isolation is reported as confounded — the model disobeys a "do not echo" instruction in 90–100% of trials — with the no-echo condition still clearly beating Base, consistent with but not proof of segmentation carrying the effect.

E INTEGER ADDITION FOLLOW-UPS

Full grids for the carry-direction (least-significant-digit-first) prompt and the scratchpad prompt on all four models. Reversed digit order does not rescue the effect on any model: Base beats every tested boundary density in 19 of 20 (model, length) combinations, with one exact tie (Qwen2.5-32B at $N{=}8$: Base 0.475 = best-density Cellular 0.475) — the qualitative conclusion is unchanged. The scratchpad follow-up is informative only at $N{=}4$ (where it agrees with the rest of the addition evidence, Cellular below Base on all four models) and is underpowered elsewhere — instruction-following on the `Final Answer:` format itself ranges from 4% (14B) to 55% (7B) across models, and both conditions collapse to near-floor by $N{=}8$–16. The carry-aligned interleaved-column placement collapses to near-zero TAR with response lengths flat at $\sim$4.2 characters regardless of N — a format-parsing failure, reported as a confounded rather than clean negative.

State-carrying checkpoints for addition. The state-carrying redesign that partially rescued global running-sum (Section 3.7) has a correctly-posed addition analogue only in least-significant-digit-first order: in the normal MSD presentation, the carry at a boundary is not yet determined by anything the model has seen, so a truthful checkpoint cannot be written there. We therefore combined LSD digit order (interleaved (a_i, b_i) pairs) with a genuine checkpoint stating the true carry bit produced by every digit pair before it — fully determined by the visible prefix, exactly like modsum's checkpoint ($N \in \{8, 16, 32, 64\}$, interval 4, 20 trials per cell, Qwen2.5-7B; the checkpoint-generation logic was independently verified against a from-scratch MSD reference implementation before any GPU run). The result is a modest, real, and insufficient improvement: the state-carrying checkpoint beats the content-free boundary at every length (e.g. 0.119 vs. 0.081 at $N{=}8$; 0.091 vs. 0.080 at $N{=}64$; gains of roughly $+0.01$ to $+0.04$ TAR), but Base (0.168–0.359) still dominates both by a wide margin throughout. The honest reading is consistent with the modsum finding while sharpening it: a real checkpoint helps in proportion to how much state there is to lose, but addition's carry chain is harder to rescue this way — plausibly because addition requires the checkpointed

state to be exactly right at every position (one wrong carry corrupts everything downstream) in a way modsum's looser dependency does not.

Carry-direction (least-significant-digit-first) prompt.

```
You are a deterministic arithmetic calculator.
Task: Add two {length}-digit numbers. Both input numbers and the expected
output are written LEAST-SIGNIFICANT-DIGIT FIRST (reversed order, i.e.
the ones digit comes first).
Constraint: The input may contain vertical bars (|) as structural
boundaries -- they carry no numeric meaning, ignore them when computing.
Output ONLY the {length}-digit sum, also least-significant-digit-first.
No spaces, no explanations.

Input: {input}
Output:
```

Model	N	Base TAR	Cellular TAR	Gap
Qwen2.5-7B	4	0.412	0.150	−0.262
Qwen2.5-7B	8	0.356	0.135	−0.221
Qwen2.5-7B	16	0.359	0.108	−0.252
Qwen2.5-7B	32	0.291	0.058	−0.232
Qwen2.5-7B	64	0.168	0.049	−0.119
Qwen2.5-14B	4	0.512	0.198	−0.315
Qwen2.5-14B	8	0.356	0.171	−0.185
Qwen2.5-14B	16	0.400	0.116	−0.284
Qwen2.5-14B	32	0.353	0.101	−0.253
Qwen2.5-14B	64	0.262	0.077	−0.185
Mistral-7B	4	0.438	0.185	−0.253
Mistral-7B	8	0.419	0.196	−0.223
Mistral-7B	16	0.369	0.120	−0.249
Mistral-7B	32	0.334	0.061	−0.273
Mistral-7B	64	0.209	0.054	−0.155
Qwen2.5-32B	4	0.463	0.247	−0.215
Qwen2.5-32B	8	0.475	0.224	−0.251
Qwen2.5-32B	16	0.425	0.119	−0.306
Qwen2.5-32B	32	0.408	0.069	−0.339
Qwen2.5-32B	64	0.338	0.054	−0.283

Table 6: LSD-reversed addition, density-averaged Cellular vs. Base per cell. (The 19-of-20 statement in the text refers to Base vs. the *best* tested density per cell, whose one exact tie is Qwen2.5-32B at N=8.)

Scratchpad prompt.

```
You are a careful arithmetic assistant.
Task: Add two {length}-digit numbers.
Constraint: The input may contain vertical bars (|) as structural
boundaries -- they carry no numeric meaning, ignore them when computing.
First, work through the addition step by step from the rightmost
(least-significant) digit to the left, tracking the carry at each step.
Then, on its own line, write exactly:
Final Answer: <the {length}-digit sum, zero-padded if needed>

Input: {input}
```

F SK FITS, BOOTSTRAP, AND MODEL SELECTION (CYCLIC SHIFT)

The full SK treatment from the earlier phase is retained here: the fitted (β_0, α) table with 500-iteration bootstrap CIs for every (model, density) condition at alphabet 2 (Table 8); the finding that well-populated Cellular densities are well-constrained (e.g. Qwen2.5-7B k=1: $\alpha = 1.037$, CI

Model	N	Base TAR	Cell. TAR	Gap	Base SAR	Cell. SAR
Qwen2.5-7B	4	0.467	0.083	-0.383	0.200	0.000
Qwen2.5-7B	8	0.058	0.072	$+0.014$	0.000	0.000
Qwen2.5-7B	16	0.217	0.019	-0.197	0.067	0.000
Qwen2.5-14B	4	0.200	0.033	-0.167	0.200	0.033
Qwen2.5-14B	8	0.000	0.017	$+0.017$	0.000	0.000
Qwen2.5-14B	16	0.013	0.000	-0.013	0.000	0.000
Mistral-7B	4	0.033	0.008	-0.025	0.000	0.000
Mistral-7B	8	0.000	0.008	$+0.008$	0.000	0.000
Mistral-7B	16	0.042	0.021	-0.021	0.000	0.000
Qwen2.5-32B	4	0.667	0.025	-0.642	0.667	0.000
Qwen2.5-32B	8	0.542	0.000	-0.542	0.400	0.000
Qwen2.5-32B	16	0.000	0.119	$+0.119$	0.000	0.044

Table 7: Scratchpad follow-up (15 trials, 3 lengths, all four addition-evaluated models). `Final Answer:` parse rates: 54.5% (7B), 4.2% (14B), 16.4% (Mistral), 18.8% (32B).

$[1.019, 1.084]$, below Base's own lower bound) while Base fits are genuinely underconstrained; AIC model selection preferring SK over one-parameter exponential decay in 16 of 22 conditions, with the 6 exceptions exactly the bound-pinned fits already flagged; length-subset sensitivity (fits robust to dropping floor-region lengths, sensitive to short-only grids); the dense near-cliff sampling analysis (helps the side with something to measure, cannot rescue the side that collapses first); the harder-alphabet bootstrap with Bonferroni correction (0 of 27 density comparisons distinguishable in either direction — "inconclusive due to data starvation," not "reverses"); the zero-width-CI degenerate case; and the upper-bound-pinned failure mode.

Binomial-likelihood MLE as a fitting robustness check. SAR(N) is a proportion of fully correct trials — a binomial observation, not a continuous measurement with Gaussian noise — so the least-squares fit on aggregated SAR points, while conventional, is not the statistically correct likelihood. We therefore refit the SK model by maximizing the binomial log-likelihood over the raw per-trial 0/1 outcomes directly, with 500-iteration bootstrap CIs and explicit flagging of bound-pinned fits. The α trends persist under the correct likelihood: for Qwen2.5-7B, Base α remains consistently higher (1.212 at alphabet 2; 1.717 at alphabet 4; 1.480 at alphabet 8) than dense-boundary α (1.051, 1.112, and 1.061 at k=1 for the three alphabets; 1.011 at alphabet-2 k=2), matching the least-squares point estimates closely where both exist (e.g. alphabet-2 Base: 1.212 under both methods; k=1: 1.037 LSQ vs. 1.051 MLE). Some k=2/k=3 cells at the larger alphabets pin at the lower bound under the MLE and are flagged as uninformative rather than read as evidence in either direction — the same convention applied to the least-squares fits throughout.

G Dose–response fit comparison, full table

The full 43-row linear/quadratic/exponential R^2 comparison (Table 9); the per-curve robustness analysis of Section 3.2 (AIC/BIC winners and leave-one-density-out CV outcomes for all 348 individual curves: quadratic preferred by AIC on 234/348, by BIC on 245/348, by held-out CV on 61/348); and the vertex-vs-N regression analysis, reported both ungated (descriptive) and CV-gated (the honest negative of Section 6: only 2 of 43 groups support a gated fit). The full 348-curve per-curve table (per-curve AIC/BIC/CV winners, fitted vertices, and bootstrap vertex CIs) is provided as `dose_response_robustness_full.csv` in the repository rather than typeset here.

H Well-formedness tables

Table 10 reports the pooled diagnostics per task, model, and condition; the by-alphabet-size/by-modulus breakdown is provided as `diagnostics_by_axis.csv` in the repository.

Model	**Method**	k	β_0	α	95% CI	N lengths	N distinct SAR
Qwen2.5-7B	Base	—	0.097	1.212	[1.017, 2.181]	7	3
Qwen2.5-7B	Cellular	1	0.005	1.037	[1.019, 1.084]	7	4
Qwen2.5-7B	Cellular	2	0.010	1.022	[1.006, 1.068]	7	5
Qwen2.5-7B	Cellular	3	0.663	1.000	[1.000, 1.679]	7	3
Qwen2.5-7B	Cellular	4	0.049	1.197	[1.197, 1.347]	6	2
Qwen2.5-14B	Base	—	0.075	1.258	[1.045, 2.236]	7	3
Qwen2.5-14B	Cellular	1	0.000	1.269	[1.199, 1.317]	7	3
Qwen2.5-14B	Cellular	2	0.000	1.153	[1.133, 1.164]	7	4
Qwen2.5-14B	Cellular	3	0.125	1.000	[1.000, 1.003]	7	5
Qwen2.5-14B	Cellular	4	0.149	1.000	[1.000, 1.323]	6	3
Qwen2.5-14B	Cellular	8	0.028	1.135	[1.111, 1.135]	5	2
Mistral-7B-v0.3	Base	—	0.035	1.781	[1.508, 2.121]	7	2
Mistral-7B-v0.3	Cellular	1	0.045	1.054	[1.005, 1.341]	7	4
Mistral-7B-v0.3	Cellular	2	0.236	1.345	[1.251, 1.552]	7	2
Mistral-7B-v0.3	Cellular	3	0.091	1.178	[1.000, 2.121]	7	3
Mistral-7B-v0.3	Cellular	4	0.062	1.246	[1.197, 1.274]	6	2
Qwen2.5-32B	Base	—	0.046	1.162	[1.032, 1.762]	7	3
Qwen2.5-32B	Cellular	1	0.000	1.199	[1.199, 1.199]	7	2
Qwen2.5-32B	Cellular	2	0.000	1.050	[1.034, 1.138]	7	4
Qwen2.5-32B	Cellular	3	0.056	1.000	[1.000, 1.111]	7	6
Qwen2.5-32B	Cellular	4	0.071	1.012	[1.000, 1.141]	6	4
Qwen2.5-32B	Cellular	8	0.028	1.135	[1.110, 1.135]	5	2

Table 8: Fitted SK parameters and 500-iteration bootstrap 95% CIs for α, alphabet 2, every (model, density) condition with a valid fit. "N distinct SAR" counts lengths producing a genuinely different mean SAR — the honest measure of how much information a fit rests on. Fits at 1.000 pin the optimizer's lower bound and are uninformative; the zero-width CI (32B, k=1) reflects zero within-length variance (SAR exactly 1 then exactly 0 at every length), not unusual confidence.

Table 10: Well-formedness diagnostics, pooled per task, model, and condition (k=3 excluded). WF = well-formed rate (trial-level AND of vocabulary validity and completion); NEO = normalized error onset.

Task	**Model**	Cond.	n	TAR	SAR	VVR	CR	WFR	TAR|WF
cyclic_shift	Mistral-7B	Base	840	0.356	0.035	0.579	0.963	0.560	0.463
cyclic_shift	Mistral-7B	Bounds	3480	0.372	0.057	0.570	0.809	0.436	0.515
cyclic_shift	Phi-3.5-instruct	Base	420	0.315	0.043	0.798	0.555	0.452	0.389
cyclic_shift	Phi-3.5-instruct	Bounds	1740	0.347	0.040	0.580	0.623	0.366	0.469
cyclic_shift	Qwen-14B	Base	420	0.465	0.071	0.433	0.974	0.410	0.659
cyclic_shift	Qwen-14B	Bounds	1740	0.627	0.197	0.493	0.780	0.375	0.846
cyclic_shift	Qwen-32B	Base	420	0.511	0.110	0.536	0.919	0.521	0.686
cyclic_shift	Qwen-32B	Bounds	1740	0.701	0.254	0.568	0.806	0.444	0.880
cyclic_shift	Qwen-7B	Base	420	0.423	0.055	0.460	0.910	0.407	0.619
cyclic_shift	Qwen-7B	Bounds	1740	0.520	0.151	0.484	0.651	0.310	0.819
character_substitution	Mistral-7B	Base	600	0.391	0.048	0.975	0.893	0.872	0.417
character_substitution	Mistral-7B	Bounds	2280	0.447	0.096	0.957	0.847	0.809	0.479
character_substitution	Phi-3.5-instruct	Base	300	0.307	0.047	0.997	0.613	0.613	0.359
character_substitution	Phi-3.5-instruct	Bounds	1140	0.317	0.055	1.000	0.739	0.739	0.343
character_substitution	Qwen-14B	Base	600	0.455	0.072	0.960	0.953	0.920	0.472
character_substitution	Qwen-14B	Bounds	2280	0.638	0.271	0.980	0.912	0.893	0.649
character_substitution	Qwen-32B	Base	600	0.482	0.095	0.962	0.867	0.840	0.521
character_substitution	Qwen-32B	Bounds	2280	0.714	0.332	0.971	0.928	0.901	0.725
character_substitution	Qwen-7B	Base	600	0.420	0.062	0.910	0.908	0.827	0.453
character_substitution	Qwen-7B	Bounds	2280	0.536	0.188	0.875	0.807	0.709	0.613
random_permutation	Mistral-7B	Base	600	0.379	0.040	0.995	0.908	0.903	0.396
random_permutation	Mistral-7B	Bounds	2280	0.423	0.078	0.978	0.833	0.815	0.449
random_permutation	Phi-3.5-instruct	Base	300	0.324	0.047	1.000	0.593	0.593	0.391
random_permutation	Phi-3.5-instruct	Bounds	1140	0.342	0.054	1.000	0.733	0.733	0.377
random_permutation	Qwen-14B	Base	600	0.445	0.070	0.988	0.973	0.962	0.450

Task	**Model**	Cond.	n	TAR	SAR	VVR	CR	WFR	TAR|WF
random_permutation	Qwen-14B	Bounds	2280	0.624	0.243	0.992	0.910	0.903	0.633
random_permutation	Qwen-32B	Base	600	0.499	0.095	0.995	0.858	0.855	0.533
random_permutation	Qwen-32B	Bounds	2280	0.720	0.327	0.994	0.923	0.918	0.725
random_permutation	Qwen-7B	Base	600	0.417	0.057	0.897	0.880	0.783	0.461
random_permutation	Qwen-7B	Bounds	2280	0.528	0.165	0.839	0.797	0.673	0.614
conditional_shift	Mistral-7B	Base	600	0.360	0.030	0.730	0.968	0.707	0.390
conditional_shift	Mistral-7B	Bounds	2280	0.394	0.050	0.729	0.857	0.607	0.447
conditional_shift	Phi-3.5-instruct	Base	300	0.347	0.033	0.813	0.550	0.440	0.432
conditional_shift	Phi-3.5-instruct	Bounds	1140	0.366	0.032	0.650	0.662	0.446	0.456
conditional_shift	Qwen-14B	Base	300	0.426	0.057	0.680	0.913	0.633	0.499
conditional_shift	Qwen-14B	Bounds	1140	0.546	0.132	0.692	0.864	0.604	0.637
conditional_shift	Qwen-32B	Base	300	0.514	0.080	0.917	0.937	0.863	0.546
conditional_shift	Qwen-32B	Bounds	1140	0.671	0.200	0.834	0.908	0.752	0.717
conditional_shift	Qwen-7B	Base	600	0.362	0.053	0.543	0.863	0.482	0.473
conditional_shift	Qwen-7B	Bounds	2280	0.384	0.050	0.332	0.829	0.290	0.563
composed_permutation	Mistral-7B	Base	400	0.179	0.000	0.995	0.910	0.905	0.186
composed_permutation	Mistral-7B	Bounds	1520	0.147	0.001	0.982	0.864	0.849	0.149
composed_permutation	Phi-3.5-instruct	Base	200	0.152	0.000	1.000	0.615	0.615	0.163
composed_permutation	Phi-3.5-instruct	Bounds	760	0.157	0.000	1.000	0.684	0.684	0.167
composed_permutation	Qwen-14B	Base	400	0.193	0.000	0.998	0.970	0.968	0.196
composed_permutation	Qwen-14B	Bounds	1520	0.182	0.003	0.999	0.909	0.908	0.187
composed_permutation	Qwen-32B	Base	400	0.162	0.000	1.000	0.860	0.860	0.168
composed_permutation	Qwen-32B	Bounds	1520	0.131	0.000	1.000	0.913	0.913	0.132
composed_permutation	Qwen-7B	Base	400	0.157	0.000	0.960	0.900	0.870	0.164
composed_permutation	Qwen-7B	Bounds	1520	0.109	0.000	0.828	0.823	0.689	0.113
copy_previous	Mistral-7B	Base	200	0.570	0.040	1.000	0.930	0.930	0.582
copy_previous	Mistral-7B	Bounds	760	0.553	0.018	0.999	0.892	0.891	0.572
copy_previous	Qwen-14B	Base	200	0.861	0.590	1.000	0.915	0.915	0.885
copy_previous	Qwen-14B	Bounds	760	0.740	0.237	0.999	0.784	0.783	0.771
copy_previous	Qwen-32B	Base	200	0.859	0.495	1.000	0.955	0.955	0.870
copy_previous	Qwen-32B	Bounds	760	0.817	0.276	1.000	0.879	0.879	0.837
copy_previous	Qwen-7B	Base	200	0.529	0.060	0.505	0.885	0.445	0.653
copy_previous	Qwen-7B	Bounds	760	0.417	0.011	0.253	0.817	0.209	0.568
running_max	Mistral-7B	Base	600	0.571	0.085	0.998	0.612	0.610	0.589
running_max	Mistral-7B	Bounds	2280	0.468	0.035	0.997	0.415	0.413	0.588
running_max	Qwen-14B	Base	600	0.922	0.735	1.000	0.993	0.993	0.924
running_max	Qwen-14B	Bounds	2280	0.849	0.438	1.000	0.868	0.868	0.870
running_max	Qwen-32B	Base	600	0.979	0.865	1.000	0.980	0.980	0.983
running_max	Qwen-32B	Bounds	2280	0.900	0.557	1.000	0.938	0.938	0.901
running_max	Qwen-7B	Base	600	0.697	0.222	0.703	0.867	0.598	0.773
running_max	Qwen-7B	Bounds	2280	0.658	0.174	0.704	0.759	0.524	0.775
running_xor	Mistral-7B	Base	200	0.431	0.000	0.775	0.830	0.610	0.466
running_xor	Mistral-7B	Bounds	760	0.465	0.011	0.862	0.714	0.601	0.522
running_xor	Phi-3.5-instruct	Base	100	0.574	0.030	1.000	0.920	0.920	0.577
running_xor	Phi-3.5-instruct	Bounds	380	0.589	0.034	1.000	0.845	0.845	0.610
running_xor	Qwen-14B	Base	200	0.748	0.180	1.000	0.895	0.895	0.769
running_xor	Qwen-14B	Bounds	760	0.716	0.066	0.997	0.925	0.922	0.729
running_xor	Qwen-32B	Base	200	0.781	0.240	1.000	0.940	0.940	0.796
running_xor	Qwen-32B	Bounds	760	0.796	0.113	1.000	0.930	0.930	0.803
running_xor	Qwen-7B	Base	200	0.411	0.010	0.880	0.585	0.500	0.462
running_xor	Qwen-7B	Bounds	760	0.401	0.007	0.524	0.662	0.363	0.516
running_sum_local	Mistral-7B	Base	600	0.321	0.005	0.753	0.787	0.572	0.390
running_sum_local	Mistral-7B	Bounds	2280	0.298	0.002	0.528	0.689	0.348	0.416
running_sum_local	Phi-3.5-instruct	Base	300	0.381	0.020	0.903	0.910	0.817	0.402
running_sum_local	Phi-3.5-instruct	Bounds	1140	0.367	0.021	0.539	0.715	0.429	0.489
running_sum_local	Qwen-14B	Base	600	0.548	0.113	0.802	0.780	0.665	0.649
running_sum_local	Qwen-14B	Bounds	2280	0.549	0.054	0.709	0.869	0.640	0.643

Task	Model	Cond.	n	TAR	SAR	VVR	CR	WFR	TAR|WF
running_sum_local	Qwen-32B	Base	600	0.590	0.117	0.830	0.900	0.765	0.660
running_sum_local	Qwen-32B	Bounds	2280	0.601	0.049	0.699	0.946	0.675	0.687
running_sum_local	Qwen-7B	Base	600	0.309	0.002	0.848	0.430	0.372	0.407
running_sum_local	Qwen-7B	Bounds	2280	0.279	0.003	0.589	0.581	0.340	0.347

I VERIFICATION STATUS

J PAULI STRING MULTIPLICATION (EXPLORATORY)

Two random length-N Pauli operator strings are multiplied position-wise, with a single accumulated global phase symbol — a hybrid local/global task. On Qwen2.5-7B the operator component never leaves floor at any tested length or density (string-correct rate 0.0 throughout), so the result is reported as exploratory and underpowered rather than as evidence in either direction.

K GEMINI PER-LENGTH AND PER-DENSITY BREAKDOWNS

Full Seq_Length $\times$ Interval breakdown tables for all nine tasks on the closed-model replication (n=3 trials per cell pooled across seeds, k=3 excluded).

L DIVIDE-AND-CONQUER AND WALL-CLOCK DETAILS

Implementation details for the decomposition baseline and the wall-clock/throughput grid across densities follow.

Exact decomposition prompt. Each chunk is scored with the identical template as the Base condition — no boundary markers, no chunk-specific instructions, no indication the input is a fragment:

```
You are a deterministic sequence transformer.
Task: Cyclic character shift.
Alphabet: A, B
Rule: Replace every character with the next one in the alphabet.
Mapping: B -> A, and A -> B.
Constraint: Output ONLY the transformed string.
No spaces, no explanations.

Input: {chunk}
Output:
```

where {chunk} is a single 4-character slice, applied $\lceil N/4 \rceil$ times per trial.

Task	**Model**	Curves	Linear R^2	Quadratic R^2	Exponential R^2
cyclic_shift	Mistral-7B	18	0.647	**0.748**	−4.5
cyclic_shift	Phi-3.5-instruct	18	0.522	**0.737**	−7.1
cyclic_shift	Qwen-14B	18	0.601	**0.909**	−2.2
cyclic_shift	Qwen-32B	18	0.554	**0.918**	−3.2
cyclic_shift	Qwen-7B	18	0.576	**0.803**	−3.1
character_substitution	Mistral-7B	12	0.772	**0.916**	−4.7
character_substitution	Phi-3.5-instruct	12	0.534	**0.722**	−9.2
character_substitution	Qwen-14B	12	0.850	**0.977**	−0.8
character_substitution	Qwen-32B	12	0.850	**0.981**	−1.2
character_substitution	Qwen-7B	12	0.789	**0.936**	−1.5
random_permutation	Mistral-7B	12	0.754	**0.932**	−5.4
random_permutation	Phi-3.5-instruct	12	0.603	**0.740**	−12.7
random_permutation	Qwen-14B	12	0.850	**0.982**	−0.9
random_permutation	Qwen-32B	12	0.849	**0.983**	−1.2
random_permutation	Qwen-7B	12	0.858	**0.957**	−1.4
conditional_shift	Mistral-7B	12	0.700	**0.895**	−3.2
conditional_shift	Phi-3.5-instruct	12	0.616	**0.839**	−3.8
conditional_shift	Qwen-14B	12	0.893	**0.938**	−0.5
conditional_shift	Qwen-32B	12	0.819	**0.948**	−3.1
conditional_shift	Qwen-7B	12	0.572	**0.736**	−7.4
composed_permutation	Mistral-7B	8	0.377	**0.882**	−12.9
composed_permutation	Phi-3.5-instruct	8	0.349	**0.557**	−12.8
composed_permutation	Qwen-14B	8	0.588	**0.818**	−31.2
composed_permutation	Qwen-32B	8	0.691	**0.855**	−3.7
composed_permutation	Qwen-7B	8	0.724	**0.878**	−5.1
running_xor	Mistral-7B	4	0.249	**0.656**	−20.1
running_xor	Phi-3.5-instruct	4	0.201	**0.383**	−95.3
running_xor	Qwen-14B	4	0.105	**0.324**	−72.5
running_xor	Qwen-32B	4	0.297	**0.453**	−59.0
running_xor	Qwen-7B	4	0.484	**0.580**	−10.9
running_sum_local	Mistral-7B	12	0.226	**0.538**	−29.7
running_sum_local	Phi-3.5-instruct	12	0.532	**0.867**	−17.5
running_sum_local	Qwen-14B	12	0.203	**0.317**	−26.1
running_sum_local	Qwen-32B	12	0.396	**0.627**	−26.2
running_sum_local	Qwen-7B	12	0.158	**0.453**	−22.2
copy_previous	Mistral-7B	4	0.204	**0.601**	−154.7
copy_previous	Qwen-14B	4	0.119	**0.887**	−35.6
copy_previous	Qwen-32B	4	0.080	**0.728**	−33.8
copy_previous	Qwen-7B	4	0.576	**0.836**	−13.8
running_max	Mistral-7B	12	0.113	**0.669**	−4.2
running_max	Qwen-14B	12	0.172	**0.842**	−38.8
running_max	Qwen-32B	12	0.145	**0.567**	−73.0
running_max	Qwen-7B	12	0.435	**0.753**	−20.9

Table 9: Aggregate dose–response fit comparison, all 43 task×model rows (mean R^2 across the row's alphabet/modulus×length curves). The quadratic (bold) wins every row; the saturating exponential is worst everywhere.

Task	7B	14B	32B	Mistral-7B	Phi-3.5-mini
`cyclic_shift`	verified	verified	verified	verified	single
`character_substitution`	verified	verified	verified	single	single
`random_permutation`	verified	verified	verified	verified	single
`conditional_shift`	verified	single	single	verified	single
`composed_permutation`	verified	verified	verified	verified	single
`running_xor`	verified	verified	verified	single	single
`running_sum_local`	verified	single	single	single	in progress
`copy_previous`	verified	single	single	single	—
`running_max`	verified	single	single	single	—

Table 11: Verification status by task and model. *Verified* = an independent, non-overlapping second batch reproduced the effect; *single* = single batch; — = not started.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	0.833	0.083	0.667	0.167	9
4	Bounds	1	1.000	0.000	1.000	0.000	9
4	Bounds	2	1.000	0.000	1.000	0.000	9
8	Base	0	0.875	0.083	0.778	0.147	9
8	Bounds	1	1.000	0.000	1.000	0.000	9
8	Bounds	2	0.944	0.030	0.667	0.167	9
8	Bounds	4	0.917	0.055	0.667	0.167	9
16	Base	0	0.729	0.065	0.111	0.111	9
16	Bounds	1	1.000	0.000	1.000	0.000	9
16	Bounds	2	0.993	0.007	0.889	0.111	9
16	Bounds	4	0.840	0.066	0.444	0.176	9
16	Bounds	8	0.688	0.067	0.000	0.000	9
32	Base	0	0.517	0.028	0.000	0.000	9
32	Bounds	1	1.000	0.000	1.000	0.000	9
32	Bounds	2	0.972	0.014	0.556	0.176	9
32	Bounds	4	0.882	0.050	0.333	0.167	9
32	Bounds	8	0.736	0.056	0.111	0.111	9
32	Bounds	16	0.708	0.040	0.000	0.000	9
64	Base	0	0.406	0.043	0.000	0.000	9
64	Bounds	1	1.000	0.000	1.000	0.000	9
64	Bounds	2	0.969	0.018	0.667	0.167	9
64	Bounds	4	0.889	0.033	0.111	0.111	9
64	Bounds	8	0.726	0.047	0.000	0.000	9
64	Bounds	16	0.608	0.047	0.000	0.000	9

Table 12: Gemini breakdown: `character_substitution`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	0.167	0.083	0.000	0.000	6
4	Bounds	1	0.500	0.158	0.167	0.167	6
4	Bounds	2	0.125	0.085	0.000	0.000	6
8	Base	0	0.125	0.046	0.000	0.000	6
8	Bounds	1	0.521	0.146	0.167	0.167	6
8	Bounds	2	0.062	0.043	0.000	0.000	6
8	Bounds	4	0.208	0.070	0.000	0.000	6
16	Base	0	0.052	0.052	0.000	0.000	6
16	Bounds	1	0.146	0.146	0.000	0.000	6
16	Bounds	2	0.000	0.000	0.000	0.000	6
16	Bounds	4	0.062	0.043	0.000	0.000	6
16	Bounds	8	0.000	0.000	0.000	0.000	6
32	Base	0	0.000	0.000	0.000	0.000	6
32	Bounds	1	0.000	0.000	0.000	0.000	6
32	Bounds	2	0.000	0.000	0.000	0.000	6
32	Bounds	4	0.000	0.000	0.000	0.000	6
32	Bounds	8	0.000	0.000	0.000	0.000	6
32	Bounds	16	0.000	0.000	0.000	0.000	6
64	Base	0	0.000	0.000	0.000	0.000	6
64	Bounds	1	0.000	0.000	0.000	0.000	6
64	Bounds	2	0.000	0.000	0.000	0.000	6
64	Bounds	4	0.000	0.000	0.000	0.000	6
64	Bounds	8	0.000	0.000	0.000	0.000	6
64	Bounds	16	0.000	0.000	0.000	0.000	6

Table 13: Gemini breakdown: `composed_permutation`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	0.917	0.059	0.778	0.147	9
4	Bounds	1	0.972	0.028	0.889	0.111	9
4	Bounds	2	0.944	0.056	0.889	0.111	9
8	Base	0	0.806	0.066	0.333	0.167	9
8	Bounds	1	1.000	0.000	1.000	0.000	9
8	Bounds	2	0.917	0.029	0.444	0.176	9
8	Bounds	4	0.819	0.073	0.444	0.176	9
16	Base	0	0.625	0.091	0.111	0.111	9
16	Bounds	1	1.000	0.000	1.000	0.000	9
16	Bounds	2	0.868	0.110	0.778	0.147	9
16	Bounds	4	0.903	0.018	0.111	0.111	9
16	Bounds	8	0.569	0.100	0.000	0.000	9
32	Base	0	0.240	0.120	0.000	0.000	9
32	Bounds	1	0.556	0.176	0.556	0.176	9
32	Bounds	2	0.552	0.175	0.444	0.176	9
32	Bounds	4	0.479	0.166	0.333	0.167	9
32	Bounds	8	0.497	0.135	0.000	0.000	9
32	Bounds	16	0.323	0.131	0.000	0.000	9
64	Base	0	0.132	0.087	0.000	0.000	9
64	Bounds	1	0.667	0.167	0.667	0.167	9
64	Bounds	2	0.111	0.111	0.111	0.111	9
64	Bounds	4	0.215	0.142	0.111	0.111	9
64	Bounds	8	0.325	0.144	0.000	0.000	9
64	Bounds	16	0.215	0.111	0.000	0.000	9

Table 14: Gemini breakdown: `conditional_shift`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	1.000	0.000	1.000	0.000	3
4	Bounds	1	1.000	0.000	1.000	0.000	3
4	Bounds	2	0.917	0.083	0.667	0.333	3
8	Base	0	1.000	0.000	1.000	0.000	3
8	Bounds	1	1.000	0.000	1.000	0.000	3
8	Bounds	2	0.917	0.042	0.333	0.333	3
8	Bounds	4	0.958	0.042	0.667	0.333	3
16	Base	0	1.000	0.000	1.000	0.000	3
16	Bounds	1	1.000	0.000	1.000	0.000	3
16	Bounds	2	0.896	0.021	0.000	0.000	3
16	Bounds	4	0.896	0.055	0.333	0.333	3
16	Bounds	8	0.958	0.021	0.333	0.333	3
32	Base	0	0.938	0.062	0.667	0.333	3
32	Bounds	1	1.000	0.000	1.000	0.000	3
32	Bounds	2	0.802	0.063	0.000	0.000	3
32	Bounds	4	0.875	0.065	0.333	0.333	3
32	Bounds	8	0.948	0.028	0.333	0.333	3
32	Bounds	16	0.979	0.010	0.333	0.333	3
64	Base	0	1.000	0.000	1.000	0.000	3
64	Bounds	1	1.000	0.000	1.000	0.000	3
64	Bounds	2	0.604	0.038	0.000	0.000	3
64	Bounds	4	0.875	0.016	0.000	0.000	3
64	Bounds	8	0.938	0.009	0.000	0.000	3
64	Bounds	16	0.990	0.005	0.333	0.333	3

Table 15: Gemini breakdown: `copy_previous`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	1.000	0.000	1.000	0.000	9
4	Bounds	1	1.000	0.000	1.000	0.000	9
4	Bounds	2	0.972	0.028	0.889	0.111	9
8	Base	0	0.944	0.030	0.667	0.167	9
8	Bounds	1	1.000	0.000	1.000	0.000	9
8	Bounds	2	0.972	0.018	0.778	0.147	9
8	Bounds	4	0.931	0.030	0.556	0.176	9
16	Base	0	0.924	0.027	0.444	0.176	9
16	Bounds	1	1.000	0.000	1.000	0.000	9
16	Bounds	2	1.000	0.000	1.000	0.000	9
16	Bounds	4	0.986	0.014	0.889	0.111	9
16	Bounds	8	0.861	0.043	0.111	0.111	9
32	Base	0	0.590	0.022	0.000	0.000	9
32	Bounds	1	1.000	0.000	1.000	0.000	9
32	Bounds	2	0.986	0.011	0.778	0.147	9
32	Bounds	4	0.941	0.015	0.222	0.147	9
32	Bounds	8	0.872	0.048	0.222	0.147	9
32	Bounds	16	0.719	0.059	0.000	0.000	9
64	Base	0	0.517	0.033	0.000	0.000	9
64	Bounds	1	1.000	0.000	1.000	0.000	9
64	Bounds	2	0.988	0.006	0.556	0.176	9
64	Bounds	4	0.929	0.026	0.333	0.167	9
64	Bounds	8	0.788	0.038	0.000	0.000	9
64	Bounds	16	0.736	0.042	0.000	0.000	9
128	Base	0	0.363	0.032	0.000	0.000	9
128	Bounds	1	0.766	0.000	0.000	0.000	9
128	Bounds	2	0.881	0.088	0.333	0.167	9
128	Bounds	4	0.862	0.057	0.222	0.147	9
128	Bounds	8	0.709	0.065	0.000	0.000	9
128	Bounds	16	0.602	0.082	0.000	0.000	9
256	Base	0	0.278	0.048	0.000	0.000	9
256	Bounds	1	0.383	0.000	0.000	0.000	9
256	Bounds	2	0.732	0.008	0.000	0.000	9
256	Bounds	4	0.833	0.052	0.222	0.147	9
256	Bounds	8	0.625	0.083	0.000	0.000	9
256	Bounds	16	0.621	0.060	0.000	0.000	9

Table 16: Gemini breakdown: `cyclic_shift`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	0.778	0.114	0.667	0.167	9
4	Bounds	1	1.000	0.000	1.000	0.000	9
4	Bounds	2	0.944	0.037	0.778	0.147	9
8	Base	0	0.792	0.078	0.444	0.176	9
8	Bounds	1	1.000	0.000	1.000	0.000	9
8	Bounds	2	0.917	0.069	0.778	0.147	9
8	Bounds	4	0.847	0.068	0.556	0.176	9
16	Base	0	0.556	0.099	0.000	0.000	9
16	Bounds	1	1.000	0.000	1.000	0.000	9
16	Bounds	2	0.993	0.007	0.889	0.111	9
16	Bounds	4	0.868	0.050	0.444	0.176	9
16	Bounds	8	0.646	0.078	0.000	0.000	9
32	Base	0	0.490	0.037	0.000	0.000	9
32	Bounds	1	1.000	0.000	1.000	0.000	9
32	Bounds	2	0.924	0.035	0.444	0.176	9
32	Bounds	4	0.778	0.058	0.333	0.167	9
32	Bounds	8	0.615	0.081	0.111	0.111	9
32	Bounds	16	0.618	0.052	0.000	0.000	9
64	Base	0	0.413	0.030	0.000	0.000	9
64	Bounds	1	1.000	0.000	1.000	0.000	9
64	Bounds	2	0.967	0.015	0.556	0.176	9
64	Bounds	4	0.762	0.075	0.000	0.000	9
64	Bounds	8	0.613	0.066	0.000	0.000	9
64	Bounds	16	0.576	0.040	0.000	0.000	9

Table 17: Gemini breakdown: `random_permutation`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	1.000	0.000	1.000	0.000	9
4	Bounds	1	1.000	0.000	1.000	0.000	9
4	Bounds	2	0.889	0.061	0.667	0.167	9
8	Base	0	1.000	0.000	1.000	0.000	9
8	Bounds	1	1.000	0.000	1.000	0.000	9
8	Bounds	2	0.722	0.090	0.333	0.167	9
8	Bounds	4	0.750	0.072	0.222	0.147	9
16	Base	0	1.000	0.000	1.000	0.000	9
16	Bounds	1	1.000	0.000	1.000	0.000	9
16	Bounds	2	0.903	0.077	0.778	0.147	9
16	Bounds	4	0.715	0.063	0.111	0.111	9
16	Bounds	8	0.785	0.072	0.333	0.167	9
32	Base	0	1.000	0.000	1.000	0.000	9
32	Bounds	1	1.000	0.000	1.000	0.000	9
32	Bounds	2	0.872	0.082	0.667	0.167	9
32	Bounds	4	0.667	0.071	0.111	0.111	9
32	Bounds	8	0.809	0.031	0.111	0.111	9
32	Bounds	16	0.906	0.036	0.333	0.167	9
64	Base	0	1.000	0.000	1.000	0.000	9
64	Bounds	1	1.000	0.000	1.000	0.000	9
64	Bounds	2	1.000	0.000	1.000	0.000	9
64	Bounds	4	0.580	0.099	0.222	0.147	9
64	Bounds	8	0.653	0.071	0.000	0.000	9
64	Bounds	16	0.854	0.034	0.000	0.000	9

Table 18: Gemini breakdown: `running_max`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	1.000	0.000	1.000	0.000	9
4	Bounds	1	0.667	0.102	0.333	0.167	9
4	Bounds	2	0.917	0.042	0.667	0.167	9
8	Base	0	0.986	0.014	0.889	0.111	9
8	Bounds	1	0.708	0.137	0.556	0.176	9
8	Bounds	2	0.722	0.040	0.000	0.000	9
8	Bounds	4	0.889	0.014	0.111	0.111	9
16	Base	0	0.972	0.011	0.556	0.176	9
16	Bounds	1	0.806	0.123	0.667	0.167	9
16	Bounds	2	0.688	0.043	0.000	0.000	9
16	Bounds	4	0.882	0.022	0.111	0.111	9
16	Bounds	8	0.958	0.010	0.333	0.167	9
32	Base	0	0.941	0.023	0.444	0.176	9
32	Bounds	1	0.528	0.147	0.333	0.167	9
32	Bounds	2	0.656	0.039	0.000	0.000	9
32	Bounds	4	0.830	0.020	0.000	0.000	9
32	Bounds	8	0.924	0.017	0.000	0.000	9
32	Bounds	16	0.948	0.016	0.222	0.147	9
64	Base	0	0.780	0.044	0.000	0.000	9
64	Bounds	1	0.866	0.097	0.222	0.147	9
64	Bounds	2	0.618	0.057	0.000	0.000	9
64	Bounds	4	0.826	0.018	0.000	0.000	9
64	Bounds	8	0.903	0.015	0.000	0.000	9
64	Bounds	16	0.920	0.014	0.000	0.000	9

Table 19: Gemini breakdown: `running_sum_local`.

N	Cond.	k	TAR	SEM	SAR	SEM	n
4	Base	0	0.917	0.083	0.667	0.333	3
4	Bounds	1	1.000	0.000	1.000	0.000	3
4	Bounds	2	1.000	0.000	1.000	0.000	3
8	Base	0	1.000	0.000	1.000	0.000	3
8	Bounds	1	1.000	0.000	1.000	0.000	3
8	Bounds	2	0.792	0.083	0.000	0.000	3
8	Bounds	4	0.917	0.042	0.333	0.333	3
16	Base	0	1.000	0.000	1.000	0.000	3
16	Bounds	1	1.000	0.000	1.000	0.000	3
16	Bounds	2	0.688	0.036	0.000	0.000	3
16	Bounds	4	0.875	0.036	0.000	0.000	3
16	Bounds	8	0.979	0.021	0.667	0.333	3
32	Base	0	0.958	0.028	0.333	0.333	3
32	Bounds	1	1.000	0.000	1.000	0.000	3
32	Bounds	2	0.802	0.055	0.000	0.000	3
32	Bounds	4	0.844	0.000	0.000	0.000	3
32	Bounds	8	0.948	0.028	0.333	0.333	3
32	Bounds	16	0.990	0.010	0.667	0.333	3
64	Base	0	0.625	0.313	0.000	0.000	3
64	Bounds	1	1.000	0.000	1.000	0.000	3
64	Bounds	2	0.750	0.009	0.000	0.000	3
64	Bounds	4	0.828	0.009	0.000	0.000	3
64	Bounds	8	0.953	0.009	0.000	0.000	3
64	Bounds	16	0.984	0.009	0.333	0.333	3

Table 20: Gemini breakdown: `running_xor`.

Method	k	N=8 (s)	N=16 (s)	N=32 (s)	N=64 (s)	Tokens/s
Base	—	0.169	0.266	0.636	1.391	∼29–31
Cellular	1	0.529	1.061	2.062	3.651	∼30–31
Cellular	2	0.257	0.515	1.019	1.985	∼31
Cellular	4	0.179	0.362	0.701	1.438	∼30–31
Cellular	8	0.155	0.309	0.556	1.102	∼30

Table 21: Per-trial wall-clock and throughput across densities (Qwen2.5-7B, single A100, per-call `time.perf_counter()`, no batching). Throughput is density-independent; total time tracks token counts, so the sparsest density is comparable to or faster than Base outright.